THEOREM B.1 Structure of an Expanded Determinant

The determinant of an $n \times n$ matrix $A = [a_{ij}]$ can be expressed as a sum of signed products, where each product contains exactly one factor from each row and exactly one factor from each column. The expansion of $\det(A)$ on any row or column also has this form.

PROOF We consider the expansion of $\det(A)$ on the first row and give a proof by induction. We have just shown that our result is true for determinants of orders 2 and 3. Let $n > 3$, and assume that our result holds for all square matrices of size smaller than $n \times n$. Let A be an $n \times n$ matrix. When we expand $\det(A)$ by minors across the first row, the only expression involving a_{1j} is $(-1)^{1+j}a_{1j}|A_{1j}|$. We apply our induction hypothesis to the determinant $|A_{1j}|$ of order $n - 1$: it is a sum of signed products, each of which has one factor from each row and column of A except for row 1 and column j. As we multiply this sum term by term by a_{1j}, we obtain a sum of products having a_{1j} as the factor from row 1 and column j, and one factor from each other row and from each other column. Thus an expression of the stated form is indeed obtained as we expand $\det(A)$ by minors across the first row.

It is clear that essentially the same argument shows that expansion across any row or down any column yields the same type of sum of signed products. ▲

Our illustration for 2×2 and 3×3 matrices indicates that we might always have the same number of products appearing in $\det(A)$ with a sign given by 1 as with a sign given by -1. This is indeed the case for determinants of order greater than 1, and the induction proof is left to the reader.

We now restate and prove Theorem 4.2.

Let $A = [a_{ij}]$ be an $n \times n$ matrix. Then

$$|A| = (-1)^{r+1}a_{r1}|A_{r1}| + (-1)^{r+2}a_{r2}|A_{r2}| + \cdots + (-1)^{r+n}a_{rn}|A_{rn}| \quad \text{(B.1)}$$

for any r from 1 to n, and

$$|A| = (-1)^{1+s}a_{1s}|A_{1s}| + (-1)^{2+s}a_{2s}|A_{2s}| + \cdots + (-1)^{n+s}a_{ns}|A_{ns}| \quad \text{(B.2)}$$

for any s from 1 to n.

PROOF OF THEOREM 4.2 We first prove Eq. (B.1) for any choice of r from 1 to n. Clearly, Eq. (B.1) holds for $n = 1$ and $n = 2$. Proceeding by induction, let $n > 2$ and assume that determinants of order less than n can be computed by using an expansion on *any row*. Let A be an $n \times n$ matrix. We show that expansion of det (A) by minors on row r is the same as expansion on row i for $i < r$. From Theorem B 1, we know that each of the expansions gives a sum of

signed products, where each product contains a single factor from each ro and from each column of A. We will compare the products containing a factors both a_{ij} and a_{rs} in each of the expansions. We consider two cases, a illustrated in Figures B.1 and B.2.

If det(A) is expanded on the ith row, the sum of signed products containin $a_{ij}a_{rs}$ is part of $(-1)^{i+j}a_{ij}|A_{ij}|$. In computing $|A_{ij}|$ we may, by our inductio assumption, expand on the rth row. For $j < s$, terms of $|A_{ij}|$ involving a_{rs} ar then $(-1)^{(r-1)+(s-1)}d$, where d is the determinant of the matrix obtained from by crossing out rows i and r and columns j and s, as shown in Figure B.1. Th exponent $(r-1)+(s-1)$ occurs because a_{rs} is in row $r-1$ and column $s-$ of A_{ij}. Thus, the part of our expansion of det(A) across the ith row that contain $a_{ij}a_{rs}$ is equal to

$$(-1)^{i+j}(-1)^{(r-1)+(s-1)}a_{ij}a_{rs}d \quad \text{for} \quad j < s. \qquad \text{(B.3}$$

For $j > s$, we consult Figure B.2 and use similar reasoning to see that th part of our expansion of det(A) across the ith row, which contains $a_{ij}a_{rs}$, i equal to

$$(-1)^{i+j}(-1)^{(r-1)+s}a_{ij}a_{rs}d \quad \text{for} \quad j > s. \qquad \text{(B.4}$$

We now expand det(A) by minors on the rth row, obtaining $(-1)^{r+s}a_{rs}|A_{rs}|$ a the portion involving a_{rs}. Expanding $|A_{rs}|$ on the ith row, using our inductio assumption, we obtain $(-1)^{i+j}a_{ij}d$ if $j < s$ and $(-1)^{i+(j-1)}a_{ij}d$ if $j > s$. Thus th part of the expansion of det(A) on the rth row, which contains $a_{ij}a_{rs}$, is equal t

$$(-1)^{r+s}(-1)^{i+j}a_{rs}a_{ij}d \quad \text{for} \quad j < s \qquad \text{(B.5}$$

or

$$(-1)^{r+s}(-1)^{i+(j-1)}a_{rs}a_{ij}d \quad \text{for} \quad j > s. \qquad \text{(B.6}$$

Expressions (B.3) and (B.5) are equal, because $(-1)^{r+s+i+j} = (-1)^{r+s+i+j-2}$ and expressions (B.4) and (B.6) are equal, because $(-1)^{r+s+i+j-1}$ is the algebrai sign of each. This concludes the proof that the expansions of det(A) by minor across rows i and r are equal.

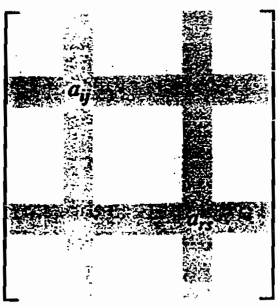


FIGURE B.1
The case $j < s$.

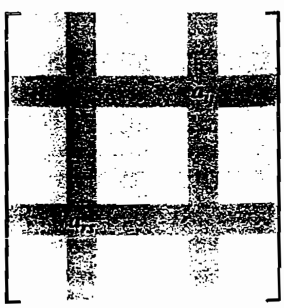


FIGURE B.2
The case $j > s$.

A similar argument shows that expansions of det(A) down columns j and s are the same.

Finally, we must show that an expansion of det(A) on a row is equal to expansion on a column. It is sufficient for us to prove that the expansion of det(A) on the first row is the same as the expansion on the first column, in view of what we have proved above. Again, we use induction and dispose of the cases $n = 1$ and $n = 2$ as trivial to check. Let $n > 2$, and assume that our result holds for matrices of size smaller than $n \times n$. Let A be an $n \times n$ matrix. Expanding det(A) on the first row yields

$$a_{11}|A_{11}| + \sum_{j=2}^{n} (-1)^{1+j} a_{1j}|A_{1j}|.$$

For $j > 1$, we expand $|A_{1j}|$ on the first column, using our induction assumption, and obtain $|A_{1j}| = \sum_{i=2}^{n} (-1)^{(i-1)+1} a_{i1} d$, where d is the determinant of the matrix obtained from A by crossing out rows 1 and i and columns 1 and j. Thus the terms in the expansion of det(A) containing $a_{1j}a_{i1}$ are

$$(-1)^{1+j+i} a_{1j} a_{i1} d. \tag{B.7}$$

On the other hand, if we expand det(A) on the first column, we obtain

$$a_{11}|A_{11}| + \sum_{i=2}^{n} (-1)^{i+1} a_{i1}|A_{i1}|.$$

For $i > 1$, expanding on the first row, using our induction assumption, shows that $|A_{i1}| = \sum_{j=2}^{n} (-1)^{1+(j-1)} a_{1j} d$. This results in

$$(-1)^{i+1+j} a_{i1} a_{1j} d$$

as the part of the expansion of det(A) containing the sum $a_{i1}a_{1j}$, and this agrees with the expression in formula (B.7). This concludes our proof. ▲

PROOF OF THEOREM 4.7 ON THE VOLUME OF AN n-BOX IN $\mathbb{R}^m$

Theorem 4.7 in Section 4.4 asserts that the volume of an n-box in $\mathbb{R}^m$ determined by independent vectors $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ can be computed as

$$\text{Volume} = \sqrt{\det(A^T A)}, \tag{B.8}$$

where A is the $m \times n$ matrix having $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ as column vectors. Before proving this result, we must first give a proper definition of the *volume* of such a box. The definition proceeds inductively. That is, we define the volume of a 1-box directly, and then we define the volume of an n-box in terms of the volume of an $(n - 1)$-box. Our definition of volume is a natural one, essentially taking the product of the measure of the base of the box and the altitude of the box. We will think of the base of the n-box determined by $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ as an

$(n-1)$-box determined by some $n-1$ of the vectors $\mathbf{a}_i$. In general, such a ba: can be selected in several different ways. We find it convenient to work wit boxes determined by *ordered* sequences of vectors. We will choose one speci: base for the box and give a definition of its volume in terms of this order of th vectors. Once we obtain the expression $\det(A^TA)$ in Eq. (B.8) for the square (the volume, we can show that the volume does not change if the order of th vectors in the sequence is changed. We will show that $\det(A^TA)$ remair unchanged if the order of the columns of A is changed.

Observe that, if an n-box is determined by $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$, then $\mathbf{a}_1$ can b uniquely expressed in the form

$$\mathbf{a}_1 = \mathbf{b} + \mathbf{p}, \qquad \text{(B.9}$$

where $\mathbf{p}$ is the projection of $\mathbf{a}_1$ on $W = \mathrm{sp}(\mathbf{a}_2, \ldots, \mathbf{a}_n)$ and $\mathbf{b} = \mathbf{a}_1 -$ is orthogonal to W. This follows from Theorem 6.1 and is illustrated in Fig ure B.3.

DEFINITION B.1 Volume of an n-Box

The volume of the 1-box determined by a nonzero vector $\mathbf{a}_1$ in $\mathbb{R}^m$ is $\|\mathbf{a}_1\|$. Let $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ be an ordered sequence of n independent vectors, and suppose that the volume of an r-box determined by an ordered sequence of r independent vectors has been defined for $r < n$. The volume of the n-box in $\mathbb{R}^m$ determined by the ordered sequence of $\mathbf{a}_i$ is the product of the volume of the "base" determined by the ordered sequence $\mathbf{a}_2, \ldots, \mathbf{a}_n$ and the length of the vector $\mathbf{b}$ given in Eq. (B.9). That is,

$$\text{Volume} = (\text{Altitude } \|\mathbf{b}\|)(\text{Volume of the base}).$$

As a first step in finding a formula for the volume of an n-box, we establist a preliminary result on determinants.

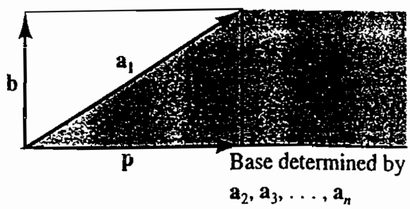


FIGURE B.3
The altitude vector b perpendicular to the base box.

THEOREM B.2 Property of $\det(A^TA)$

Let $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ be vectors in $\mathbb{R}^m$, and let A be the $m \times n$ matrix with jth column vector $\mathbf{a}_j$. Let B be the $m \times n$ matrix obtained from A by replacing the first column of A by the vector

$$\mathbf{b} = \mathbf{a}_1 - r_2\mathbf{a}_2 - \cdots - r_n\mathbf{a}_n$$

for scalars $r_2, \ldots, r_n$. Then

$$\det(A^TA) = \det(B^TB). \tag{B.10}$$

PROOF The matrix B can be obtained from the matrix A by a sequence of $n - 1$ elementary *column-addition* operations. Each of the elementary column operations can be performed on A by multiplying A on the *right* by an elementary matrix formed by executing the same elementary column-addition operation on the $n \times n$ identity matrix I. Each elementary column-addition matrix therefore has the same determinant 1 as the identity matrix I. Their product is an $n \times n$ matrix E such that $B = AE$, and $\det(E) = 1$. Using properties of determinants and the transpose operation, we have

$$\begin{aligned}\det(B^TB) &= \det((AE)^T(AE)) = \det(E^T(A^TA)E)\\ &= 1 \cdot \det(A^TA) \cdot 1 = \det(A^TA).\end{aligned}$$ ▲

We can now prove our volume formula in Theorem 4.7.

> The volume of the n-box in $\mathbb{R}^m$ determined by the ordered sequence $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ of n independent vectors is given by
>
> $$\text{Volume} = \sqrt{\det(A^TA)},$$
>
> where A is the $m \times n$ matrix with $\mathbf{a}_j$ as jth column vector.

PROOF OF THEOREM 4.7 Because our volume was defined inductively, we give an inductive proof. The theorem is valid if $n = 1$ or 2, by Eqs. (1) and (2), respectively, in Section 4.4. Let $n > 2$, and suppose that the theorem is proved for all k-boxes for $k \leq n - 1$. If we write $\mathbf{a}_1 = \mathbf{b} + \mathbf{p}$, as in Eq. (B.9), then, because $\mathbf{p}$ lies in $\text{sp}(\mathbf{a}_2, \ldots, \mathbf{a}_n)$, we have

$$\mathbf{p} = r_2\mathbf{a}_2 + \cdots + r_n\mathbf{a}_n$$

for some scalars $r_2, \ldots, r_n$, so

$$\mathbf{b} = \mathbf{a}_1 - \mathbf{p} = \mathbf{a}_1 - r_2\mathbf{a}_2 - \cdots - r_n\mathbf{a}_n.$$

Let B be the matrix obtained from A by replacing the first column vector $\mathbf{a}_1$ of A by the vector $\mathbf{b}$, as in Theorem B.2. Because $\mathbf{b}$ is orthogonal to each of the vectors $\mathbf{a}_2, \ldots, \mathbf{a}_n$, which determine the base of our box, we obtain

$$B^TB = \begin{bmatrix} \mathbf{b}\cdot\mathbf{b} & 0 & \cdots & 0 \\ 0 & \mathbf{a}_2\cdot\mathbf{a}_2 & \cdots & \mathbf{a}_2\cdot\mathbf{a}_n \\ \vdots & \vdots & & \vdots \\ 0 & \mathbf{a}_n\cdot\mathbf{a}_2 & \cdots & \mathbf{a}_n\cdot\mathbf{a}_n \end{bmatrix}. \tag{B.11}$$

From Eq. (B.11), we see that

$$\det(B^TB) = \|\mathbf{b}\|^2 \begin{vmatrix} \mathbf{a}_2\cdot\mathbf{a}_2 & \cdots & \mathbf{a}_2\cdot\mathbf{a}_n \\ \vdots & & \vdots \\ \mathbf{a}_n\cdot\mathbf{a}_2 & \cdots & \mathbf{a}_n\cdot\mathbf{a}_n \end{vmatrix}.$$

By our induction assumption, the square of the volume of the $(n-1)$-box in $\mathbb{R}^m$ determined by the ordered sequence $\mathbf{a}_2, \ldots, \mathbf{a}_n$ is

$$\begin{vmatrix} \mathbf{a}_2\cdot\mathbf{a}_2 & \cdots & \mathbf{a}_2\cdot\mathbf{a}_n \\ \vdots & & \vdots \\ \mathbf{a}_n\cdot\mathbf{a}_2 & \cdots & \mathbf{a}_n\cdot\mathbf{a}_n \end{vmatrix}.$$

Applying Eq. (B.10) in Theorem B.2, we obtain

$$\det(A^TA) = \det(B^TB) = \|\mathbf{b}\|^2(\text{Volume of the base})^2 = (\text{Volume})^2.$$

This proves our theorem. ▲

COROLLARY Independence of Order

The volume of a box determined by the independent vectors $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ and defined in Definition B.1 is independent of the order of the vectors; in particular, the volume is independent of a choice of base for the box.

PROOF A rearrangement of the sequence $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ of vectors corresponds to the same rearrangement of the columns of matrix A. Such a rearrangement of the columns of A can be accomplished by multiplying A on the right by a product of $n \times n$ elementary *column-interchange* matrices, having determinant ± 1. As in the proof of Theorem B.2, we see that, for the resulting matrix $B = AE$, we have $\det(B^TB) = \det(A^TA)$ because $\det(E^T)\det(E) = 1$. ▲

C LINTEK ROUTINES

Below are listed the names and brief descriptions of the routines that make up the computer software LINTEK designed for this text.

VECTGRPH	Gives graded quizzes on vector geometry based on displayed graphics. Useful throughout the course.
MATCOMP	Performs matrix computations, solves linear systems, and finds real eigenvalues and eigenvectors. For use throughout the course.
YUREDUCE	Enables the user to select items from a menu for step-by-step row reduction of a matrix. For Sections 1.4–1.6, 9.2, and 10.3.
EBYMTIME	Gives a lower bound for the time required to find the determinant of a matrix using only repeated expansion by minors. For Section 4.2.
ALLROOTS	Provides step-by-step execution of Newton's method to find both real and complex roots of a polynomial with real or complex coefficients; may be used in conjunction with MATCOMP to find complex as well as real eigenvalues of a small matrix. For Section 5.1.
QRFACTOR	Executes the Gram–Schmidt process, gives the QR-factorization of a suitable matrix A, and can be used to find least-squares solutions. For Section 6.2. Also, for Section 8.4, step-by-step computation by the QR-algorithm of both real and complex eigenvalues of a matrix. The user specifies shifts and when to decouple.
YOUFIT	The user can experiment with a graphic to try to find the least-squares linear, quadratic, or exponential fit of two-dimensional data. The computer can then be asked to find the best fit. For Section 6.5.
POWER	Executes, one step at a time, the power method (with deflation) for finding eigenvalues and eigenvectors. For Section 8.4.

JACOBI	Executes, one "rotation" at a time, the method of Jacobi for diagonalizing a symmetric matrix. For Section 8.4.
TIMING	Times algebraic operations and flops; also times various methods for solving linear systems. For Sections 10.1 and 10.2.
LUFACTOR	Gives the factorizations $A = LU$ or $PA = LU$, which can then be used to solve $A\mathbf{x} = \mathbf{b}$. For Section 10.2.
HILBERT	Enables the user to experiment with ill-conditioned Hilbert matrices. For Section 10.3.
LINPROG	A graphics program allowing the user to estimate solutions of two-variable linear programming problems. For use with the previous edition of our text.
SIMPLEX	Executes the simplex method with optional display of tableau to solve linear programming problems. For use with Chapter 10 of the previous edition of our text.

D MATLAB PROCEDURES AND COMMANDS USED IN THE EXERCISES

The MATLAB exercises in our text illustrate and facilitate computations in linear algebra. Also, they have been carefully designed, starting with those in Section 1.1, to introduce students gradually to MATLAB, explaining procedures and commands as the occasion for them arises. It is not necessary to study this appendix before starting right in with the MATLAB exercises in Section 1.1. In case a student has forgotten a procedure or command, we summarize here for easy reference the ones needed for the exercises. The information given here is only a small fraction of that available from the MATLAB manual, which is of course the best reference.

GENERAL INFORMATION

MATLAB prints $\gg$ on the screen as its *prompt* when it is ready for your next command.

MATLAB is *case sensitive*—that is, if you have set $n = 3$, then N is undefined. Thus you can set X equal to a 3×2 matrix and x equal to a row vector.

To view on the screen the value, vector, or matrix currently assigned to a variable such as A or x, type the variable and press the Enter key.

For information on a command, enter **help** followed by a space and the name of the command or function. For example, entering **help** $*$ will bring the response that $X * Y$ is the matrix product of X and Y, and entering **help eye** will inform us that eye(n) is the $n \times n$ identity matrix.

The *most recent* commands you have given MATLAB are kept in a *stack*, and you can move back and forth through them using the up-arrow and down-arrow keys. Thus if you make a mistake in typing a command and get an error message, you can correct the error by striking the up-arrow key to put the command by the cursor and then edit it to correct the error, avoiding retyping the entire command. The exercises frequently ask you to execute a command you recently gave, and you can simply use the up-arrow key until the command is at the cursor, and then press the Enter key to execute it again. This saves a lot of typing. If you know you will be using a command again, don't let it get too far back before asking to execute it again, or it may have been removed from the command stack. Executing it puts it at the top of the stack.

DATA ENTRY

Numerical, vector, and matrix data can be entered by the variable name for the data followed by an equal sign and then the data. For example, entering **x = 3** assigns the value 3 to the letter x, and then entering **y = sin(x)** will then assign to y the value sin(3). Entering **y = sin(x)** before a value has been assigned to x will produce an error message. Values must have been assigned to a variable before the variable can be used in a function or computation. If you do not assign a name to the result of a computation, then it is assigned the temporary name "ans." Thus this reserved name *ans* should not be used in your work because its value will be changed at the next computation having no assigned name.

The constant $\pi = 4\tan^{-1}(1)$ can be entered as **pi**.

Data for a vector or a matrix should be entered between square brackets with a space between numbers and with rows separated by a semicolon. For example, the matrix

$$A = \begin{bmatrix} 2 & -1 & 3 & 6 \\ 0 & 5 & 12 & -7 \\ 4 & -2 & 9 & 11 \end{bmatrix}$$

can be entered as

A = [2 −1 3 6; 0 5 12 −7; 4 −2 9 11],

which will produce the response

$A =$

$$\begin{matrix} 2 & -1 & 3 & 6 \\ 0 & 5 & 12 & -7 \\ 4 & -2 & 9 & 11 \end{matrix}$$

from MATLAB. If you do not wish your data entry to be *echoed* in this fashion for proofreading, put a semicolon after your data entry line. The semicolon can also be used to separate several data items or commands all entered on one line. For example, entering

x = 3; sin(x); v = [1 −3 4]; C = [2 −1; 3 5];

will result in the variable assignments

$$x = 3 \text{ and } ans = \sin(3) \text{ and } \mathbf{v} = [1, -3, 4] \text{ and } C = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix}$$

and no data will be echoed. If the final semicolon were omitted, the data for the matrix C would be echoed. If you run out of space on a line and need to continue data on the next line, type at least two periods .. and then immediately press Enter and continue entry on the next line.

Matrices can be glued together to form larger matrices using the same form of matrix entry, provided the gluing will still produce a rectangular array. For example, if A is a 3×4 matrix, B is 3×5, and C is 2×4, then entering **D = [A B]** will produce a 3×9 matrix D consisting of the four columns of A followed at the right by the five columns of B, whereas entering **E = [A ; C]** will produce a 5×4 matrix consisting of the first three rows of A followed below by the two rows of C.

NUMERIC OPERATIONS

Use + for addition, − for subtraction, * for multiplication, / for division, and ^ for exponentiation. Thus entering **6^3** produces the result ans = 216.

MATRIX OPERATIONS AND NOTATIONS

Addition, subtraction, multiplication, division, and exponentiation are denoted just as for numeric data, always assuming that they are defined for the matrices specified. For example, if A and B are square matrices of the same size, then we can form $A + B$, $A - B$, $A * B$, $2 * A$, and A^4. If A and B are invertible, then MATLAB interprets A/B as AB^{-1} and interprets $A\backslash B$ as $A^{-1}B$.

A period before an operation symbol, such as .* or .^, indicates that the operation is to be performed at each corresponding position i,j in a matrix rather than on the matrix as a whole. Thus entering **C = A .* B** produces the matrix C such that $c_{ij} = a_{ij}b_{ij}$, and entering **D = A.^2** produces the matrix D where $d_{ij} = a_{ij}^2$.

The transpose of A is denoted by A', so that entering **A = A′** will replace A by its transpose. Entering **b = [1; 2; 3]** or entering **b = [1 2 3]′** creates the same column vector **b**.

Let A be a 3×5 matrix in MATLAB. Entries are numbered, starting from the upper left corner and proceding down each column in turn. Thus $A(6)$ is the last entry in the second column of A. This entry is also identified as $A(3,2)$; we think of the 3 and 2 as subscripts as in our text notation a_{32}. Use of a colon between two integers has the meaning of "through." Thus, entering **v = A(2,1:4)** will produce a row vector **v** consisting of entries 1 through 4 (the first four components) of the second row vector of A, whereas entering **B = A(1:2,2:4)** will produce the 2×3 submatrix of A consisting of columns 2, 3, and 4 of rows 1 and 2. The colon by itself in place of a subscript has the meaning of "all," so that entering **w = A(:,3)** sets **w** equal to the third column vector of A. For another example, if **x** is a row vector with five components, then entering **A(2,:) = x** will replace the second row vector of A by **x**.

The colon can be used in the "through" sense to enter vectors with components incremented by a specified amount. Entering **y = 4 : −.5 : 2** sets **y** equal to the vector [4, 3.5, 3, 2.5, 2] with increment −0.5, whereas entering **x = 1 : 6** will set **x** equal to the vector [1, 2, 3, 4, 5, 6] with default increment 1 between components.

MATLAB COMMANDS REFERRED TO IN THIS TEXT

Remember that all variables must have already been assigned numeric, vector, or matrix values. Following is a summary of the commands used in the MATLAB exercises of our text, together with a brief description of what each one does.

acos(x)	returns the arccosine of every entry in (the number, vector, or matrix) x.
bar(v)	draws a bar graph wherein the height of the ith bar is the ith component of the vector **v**.
clear	erases all variables; all data are lost.
clear A w x	erases the variables A, w, and x only.
clock	returns the vector [year month day hour minute second]
det(A)	returns the determinant of the square matrix A.
eig(A)	returns a column vector whose components are the eigenvalues of A. Entering **[V, D] = eig(A)** creates a matrix V whose column vectors are eigenvectors of A and a diagonal matrix D whose jth entry on the diagonal is the eigenvalue associated with the jth column of V, so that $AV = VD$.
etime(t1,t0)	returns the elapsed time between two clock output vectors **t0** and **t1**. If we set **t0 = clock** before executing a computation, then the command **etime(clock,t0)** immediately after the computation gives the time required for the computation.
exit	leaves MATLAB and returns us to DOS.
eye(n)	returns the $n \times n$ identity matrix.
for i = 1:n, . . . ; end	causes the routine placed in the . . . portion to be executed n times with the value of i incremented by 1 each time. The value of n must not exceed 1024 in the student version of MATLAB.
format long	causes data to be printed in scientific format with about 16 significant digits until we specify otherwise. (See **format short.**)
format short	causes data to be printed with about five significant digits until we specify otherwise. (See **format long.**) The default format when MATLAB is accessed is the short one.
help	produces a list of topics for which we can obtain an on-screen explanation. To obtain help on a specific command or function—say, the det function—enter **help det**.

hilb(n)	returns the $n \times n$ Hilbert matrix.
invhilb(n)	returns the result of an attempted computation of the inverse of the $n \times n$ Hilbert matrix.
inv(A)	returns the inverse of the matrix A.
log(x)	returns the natural logarithm of every entry in (the number, vector, or matrix) x.
lu	The command **[L, U, P] = lu(A)** returns a lower-triangular matrix L, an upper-triangular matrix U, and a permutation matrix P such that $PA = LU$.
max(x)	returns the maximum of the components of the vector **x**.
max(X)	returns a row vector whose jth component is the maximum of the components of the jth column vector of X.
mean(x)	returns the arithmetic mean of the components of the vector **x**. That is, if **x** has n components, then mean(**x**) $= (x_1 + x_2 + \cdots + x_n)/n$.
mean(A)	returns the row vector whose jth component is the arithmetic mean of the jth column of the matrix A. (See **mean(x)**.) If A is an $m \times n$ matrix, then mean(mean(A)) = (the sum of all mn entries)/(mn).
mesh(A)	draws a three-dimensional perspective mesh surface whose height above the point (i,j) in a horizontal reference plane is a_{ij}. Let x be a vector of m components and **y** a vector of n components. The command **[X,Y] = meshdom(x, y)** creates matrices X and Y where matrix X is an array of x-coordinates of all the mn points (x_i, y_j) over which we want to plot heights for our mesh surface, and the matrix Y is a similar array of y-coordinates. See the explanation preceding Exercise M10 in Section 1.3 for a description of the use of **mesh** and **meshdom** to draw a surface graph of a function $z = f(x, y)$.
min(x)	returns the minimum of the components of the vector **x**.
min(X)	returns a row vector whose jth component is the minimum of the components of the jth column vector of X.
norm(x)	returns the usual norm $\lVert \mathbf{x} \rVert$ of the vector **x**.
ones(m,n)	returns the $m \times n$ matrix with all entries equal to 1.
ones(n)	returns the $n \times n$ matrix with all entries equal to 1.
piot(x)	draws the polygonal approximation graph of a function f such that $f(i) = x_i$ for a vector **x**.

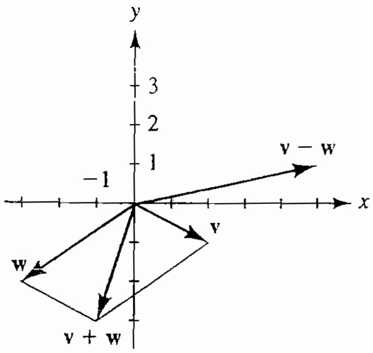
plot(x, y)	draws the polygonal approximation graph of a function f such that $f(x_i) = y_i$ for vectors **x** and **y**.
poly(A)	returns a row vector whose components are the coefficients of the characteristic polynomial $\det(\lambda I - A)$ of the square matrix A. Entering **p = poly(A)** assigns this vector to the variable **p**. (See **roots(p)**.)
qr	The command **[Q, R] = qr(A)** returns an orthogonal matrix Q and an upper-triangular matrix R such that $A = QR$. If matrix A is an $m \times n$ matrix, then R is also $m \times n$ whereas Q is $m \times m$.
quit	leaves MATLAB and returns us to DOS.
rand(m,n)	returns an $m \times n$ matrix with random number entries between 0 and 1.
rand(n)	returns an $n \times n$ matrix with random number entries between 0 and 1.
rat(x, 's')	causes the vector (or matrix) **x** to be printed on the screen with entries replaced by rational (fractional) approximations. The entries of **x** are not changed. Hopefully, entries that should be equal to fractions with small denominators will be replaced by those fractions.
roots(p)	returns a column vector of roots of the polynomial whose coefficients are in the row vector **p**. (See **poly(A)**.)
rot90(A)	returns the matrix obtained by "rotating the matrix A" counterclockwise 90°. Entering **B = rot90(A)** produces this matrix B.
rref(A)	returns the reduced row-echelon form of the matrix A.
rrefmovie(A)	produces a "movie" of reduction of A to reduced row-echelon form. It may go by so fast that it is hard to study, even using the *Pause* key. See the comment following Exercise M9 in Section 1.4 about fixing this problem.
size(A)	returns the two-component row vector $[m, n]$, where A is an $m \times n$ matrix.
stairs(x)	draws a stairstep graph of the vector **x**, where the height of the ith step is x_i.
sum(x)	returns the sum of all the entries in the vector **x**.
triu(A)	returns the upper-triangular matrix obtained by replacing all entries a_{ij} for $j > i$ in A by zero.
who	creates a list on the screen of all variables that are currently assigned numeric, vector, or matrix values.
zeros(m,n)	returns the $m \times n$ matrix with all entries equal to 0.
zeros(n)	returns the $n \times n$ matrix with all entries equal to 0.

NSWERS TO MOST ODD-
UMBERED EXERCISES

APTER 1

tion 1.1

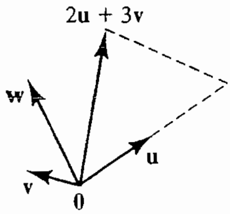
1. $\mathbf{v} + \mathbf{w} = [-1, -3]$
$\mathbf{v} - \mathbf{w} = [5, 1]$



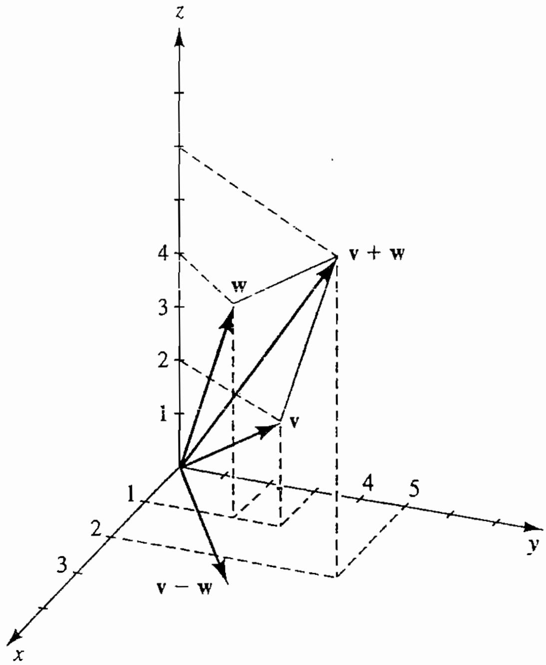
5. $[-11, 9, -4]$ 7. $[6, 4, -5]$

9. $[17, -18, 3, -20]$ **11.** $[20, -12, 6, -20]$

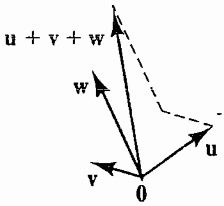
3.



3. $\mathbf{v} + \mathbf{w} = 2\mathbf{i} + 5\mathbf{j} + 6\mathbf{k}$
$\mathbf{v} - \mathbf{w} = \mathbf{j} - 2\mathbf{k}$



15.



17. $r \approx 1.5$; $s \approx 1.8$ 19. $r \approx .5$, $s \approx -1.2$

21. -1 23. $-\frac{2}{5}$ 25. All $c \neq 15$

27. 0 29. -27 31. $5\mathbf{i} - \mathbf{j}$

33. $[1, -3, -4, 13]$

35. $x\begin{bmatrix}3\\1\\2\end{bmatrix} + y\begin{bmatrix}-2\\-1\\1\end{bmatrix} + z\begin{bmatrix}4\\-3\\-5\end{bmatrix} = \begin{bmatrix}10\\0\\-3\end{bmatrix}$

37. **a.**
$$\begin{aligned} -3p \phantom{{}-2q} - 4r + 6s &= 8 \\ 4p - 2q + 3r \phantom{{}+7s} &= -3 \\ 6p + 5q - 2r + 7s &= 1 \end{aligned}$$

b. $p\begin{bmatrix}-3\\4\\6\end{bmatrix} + q\begin{bmatrix}0\\-2\\5\end{bmatrix} + r\begin{bmatrix}-4\\3\\-2\end{bmatrix} + s\begin{bmatrix}6\\0\\7\end{bmatrix} = \begin{bmatrix}8\\-3\\1\end{bmatrix}$

39. F T F F T F T F F T

M1. $[-58, 79, -36, -190]$

M3. Error using + because a and u have different numbers of components.

M5. **a.** $[-2.4524, 4.4500, -11.3810]$

b. $[-2.45238095238095,$ $4.45000000000000,$ $-11.38095238095238]$

c. $\left[-\frac{103}{42}, \frac{89}{20}, -\frac{293}{21}\right]$

M7. **a.** $[0.0357, 0.0075, 0.1962]$

b. $[0.03571428571429,$ $0.00750000000000, 0.19619047619048]$

c. $\left[\frac{1}{28}, \frac{3}{400}, \frac{103}{525}\right]$

M9. Error using + because $\mathbf{u}$ is a row vector and $\mathbf{u}^T$ is a column vector.

Section 1.2

1. $\sqrt{26}$ 3. $\sqrt{26}$ 5. $\sqrt{478}$

7. $\frac{1}{\sqrt{26}}[-1, 3, 4]$ 9. -3 11. 3

13. $\cos^{-1}\frac{11}{\sqrt{364}} \approx 54.8°$ 15. $-\frac{43}{3}$

17. $[13, -5, 7]$ or any nonzero multiple of it

19. $15\sqrt{6}$ 21. -540

25. Perpendicular

27. Parallel with opposite direction

29. Neither

31. The vector $\mathbf{v} - \mathbf{w}$, when translated to start at $(w_1, w_2, \ldots, w_n)$, reaches to the tip of the vector $\mathbf{v}$, so its length is the distance from $(w_1, w_2, \ldots, w_n)$ to $(v_1, v_2, \ldots, v_n)$.

33. $\sqrt{33}$ 35. 10 37. $\frac{100}{\sqrt{3}}$ lb

39. **b.** $\mathbf{F}_2 = -T_2(\sin\theta_2)\mathbf{i} + T_2(\cos\theta_2)\mathbf{j}$

c. $T_1(\sin\theta_1) - T_2(\sin\theta_2) = 0$, $T_1(\cos\theta_1) + T_2(\cos\theta_2) = 100$

d. $T_1 = \frac{100\sqrt{2}}{\sqrt{3}+1}$ lb, $T_2 = \frac{200}{\sqrt{3}+1}$ lb

M1. $\|\mathbf{a}\| \approx 6.3246$; the two results agree.

M3. $\|\mathbf{u}\| \approx 1.8251$

M5. **a.** 485.1449

b. Not found by MATLAB

M7. Angle ≈ 0.9499 radians

M9. Angle $\approx 147.4283°$

Section 1.3

1. $\begin{bmatrix}-6 & 3 & 9\\12 & 0 & -3\end{bmatrix}$ 3. $\begin{bmatrix}2 & 2 & 1\\9 & -1 & 2\end{bmatrix}$

5. $\begin{bmatrix}6 & -3\\-3 & 1\\-2 & 5\end{bmatrix}$ 7. Impossible

9. $\begin{bmatrix}-130 & 140\\110 & -60\end{bmatrix}$ 11. Impossible

13. $\begin{bmatrix}27 & -35\\-4 & 26\end{bmatrix}$ 15. $\begin{bmatrix}20 & -2 & -10\\-2 & 1 & 3\\-10 & 3 & 10\end{bmatrix}$

17. a. $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **b.** $\begin{bmatrix} 128 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

19. $xy = [-14]$, $yx = \begin{bmatrix} -8 & 12 & -4 \\ 2 & -3 & 1 \\ -6 & 9 & -3 \end{bmatrix}$

21. T F F T F T T T F T

23. $(A\mathbf{b})^T(A\mathbf{c}) = \mathbf{b}^T A^T A\mathbf{c}$

27. The (i, j)th entry in $(r + s)A$ is $(r + s)a_{ij} = ra_{ij} + sa_{ij}$, which is the (i, j)th entry in $rA + sA$. Because $(r + s)A$ and $rA + sA$ have the same size, they are equal.

33. Let $A = [a_{ij}]$ be an $m \times n$ matrix, $B = [b_{jk}]$ be an $n \times r$ matrix, and $C = [c_{kq}]$ be an $r \times s$ matrix. Then the ith row of AB is

$$\left[\sum_{j=1}^{n} a_{ij}b_{j1}, \sum_{j=1}^{n} a_{ij}b_{j2}, \ldots, \sum_{j=1}^{n} a_{ij}b_{jr}\right],$$

so the (i, q)th entry in $(AB)C$ is

$$\left(\sum_{j=1}^{n} a_{ij}b_{j1}\right)c_{1q} + \left(\sum_{j=1}^{n} a_{ij}b_{j2}\right)c_{2q} + \cdots + \left(\sum_{j=1}^{n} a_{ij}b_{jr}\right)c_{rq} = \sum_{k=1}^{r}\left(\sum_{j=1}^{n} (a_{ij}b_{jk}c_{kq})\right).$$

Further, the qth column of BC has components

$$\sum_{k=1}^{r} b_{1k}c_{kq}, \sum_{k=1}^{r} b_{2k}c_{kq}, \ldots, \sum_{k=1}^{r} b_{nk}c_{kq},$$

so the (i, q)th entry in $A(BC)$ is

$$a_{i1}\left(\sum_{k=1}^{r} b_{1k}c_{kq}\right) + a_{i2}\left(\sum_{k=1}^{r} b_{2k}c_{kq}\right) + \cdots + a_{in}\left(\sum_{k=1}^{r} b_{nk}c_{kq}\right) = \sum_{j=1}^{n} a_{ij}\left(\sum_{k=1}^{r} b_{jk}c_{kq}\right) = \sum_{j=1}^{n}\left(\sum_{k=1}^{r} (a_{ij}b_{jk}c_{kq})\right) = \sum_{k=1}^{r}\left(\sum_{j=1}^{n} (a_{ij}b_{jk}c_{kq})\right).$$

Because $(AB)C$ and $A(BC)$ are both $m \times s$ matrices with the same (i, q)th entry, they must be equal.

35. a. $n \times m$ **b.** $n \times n$ **c.** $m \times m$

39. Because $(AA^T)^T = (A^T)^T A^T = AA^T$, we see that AA^T is symmetric.

41. a. The jth entry in column vector $A\mathbf{e}_j$ is $[a_{i1}, a_{i2}, \cdots, a_{in}] \cdot \mathbf{e}_j = a_{ij}$. Therefore, $A\mathbf{e}_j$ is the jth column vector of A.

b. (i) We have $A\mathbf{e}_j = \mathbf{0}$ for each $j = 1, 2, \ldots, n$; so, by part a, the jth column of A is the zero vector for each j. That is, $A = O$. (ii) $A\mathbf{x} = B\mathbf{x}$ for each $\mathbf{x}$

if and only if $(A - B)\mathbf{x} = \mathbf{0}$ for each $\mathbf{x}$,

if and only if $A - B = O$ by part (i),

if and only if $A = B$.

45. These matrices do not commute for any value of r.

47. 2989 **49.** 348650

51. 32 **53.** -41558

M1. a. 32 **b.** 14629 **c.** -41558

M3. $\begin{bmatrix} 141 & -30 & -107 \\ -30 & 50 & 18 \\ -107 & 18 & 189 \end{bmatrix}$ **M5.** -117

M7. The expected mean is approximately 0.5. Your experimental values will differ from ours, and so we don't give ours.

Section 1.4

1. a. $\begin{bmatrix} 1 & 3 & 2 \\ 0 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix}$; **b.** $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

3. a. $\begin{bmatrix} -1 & 1 & 2 & 0 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$; **b.** $\begin{bmatrix} 1 & 0 & 0 & 1.5 \\ 0 & 1 & 0 & 1.5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

5. a. $\begin{bmatrix} 1 & -3 & 0 & 0 & -1 \\ 0 & 0 & 1 & 3 & -4 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 16 \end{bmatrix}$;

b. $\begin{bmatrix} 1 & -3 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

7. $\mathbf{x} = \begin{bmatrix} 5 - 6r \\ 2 - 4r \\ r \end{bmatrix}, \begin{bmatrix} -7 \\ -6 \\ 2 \end{bmatrix}$

9. $\mathbf{x} = \begin{bmatrix} 1-2r \\ -2-r-3s \\ r \\ s \end{bmatrix}, \begin{bmatrix} -5 \\ 1 \\ 3 \\ -2 \end{bmatrix}$

11. $\mathbf{x} = \begin{bmatrix} 0 \\ 2 \\ -5 \\ 2 \end{bmatrix}$

13. $x = 2, y = -4$

15. $x = -3, y = 2, z = 4$

17. $\mathbf{x} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$

19. Inconsistent

21. $\mathbf{x} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

23. $\mathbf{x} = \begin{bmatrix} -8 \\ -23-5s \\ -7+s \\ 2s \end{bmatrix}$

25. Yes

27. No

29. F F T T F T T T F T

31. $x_1 = -1, x_2 = 3$

33. $x_1 = 2, x_2 = -3$

35. $x_1 = 1, x_2 = -1, x_3 = 1, x_4 = 2$

37. $x_1 = -3, x_2 = 5, x_3 = 2, x_4 = -3$

39. All b_1 and b_2

41. All b_1, b_2, b_3 such that $b_3 + b_2 - b_1 = 0$

43. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

45. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

47. $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

49. $\begin{bmatrix} 1 & -60 & 0 & -15 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -12 & 0 & -3 \end{bmatrix}$

51. $\begin{bmatrix} 2 & -30 & 5 & -10 \\ -4 & 121 & -20 & 40 \\ 0 & -6 & 1 & -2 \\ 0 & 3 & 0 & 1 \end{bmatrix}$

57. $a = 1, b = -2, c = 1, d = 2$

59. $\mathbf{x} = \begin{bmatrix} -1 \\ 7 \end{bmatrix}$

61. $\mathbf{x} \approx \begin{bmatrix} -1.2857 \\ 3.1429 \\ 1.2857 \end{bmatrix}$

63. $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \\ 2 \end{bmatrix}$

65. $\begin{bmatrix} 1 & 2 & 0 & 0 & 3 & -6 \\ 0 & 0 & 1 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 & -1 & 3 \end{bmatrix}$

67. $\mathbf{x} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$

M1. $\mathbf{x} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

M3. Inconsistent

M5. $\mathbf{x} \approx \begin{bmatrix} 1-11s \\ 3-7s \\ s \end{bmatrix}$

M7. $\mathbf{x} = \begin{bmatrix} -13-2r+14s \\ r \\ -5+5s \\ s \end{bmatrix}$

M9. $\mathbf{x} \approx \begin{bmatrix} 0.0345 \\ -0.5370 \\ -1.6240 \\ 0.1222 \\ 0.8188 \end{bmatrix}$

Section 1.5

1. a. $A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

b. The matrix A itself is an elementary matrix.

3. Not invertible

5. a. $A^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(Other answers are possible.)

7. a. $A^{-1} = \begin{bmatrix} -7 & 5 & 3 \\ 3 & -2 & -2 \\ 3 & -2 & -1 \end{bmatrix}$;

b. $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$

9. $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{5} \end{bmatrix}$

11. The span of the column vectors is $\mathbb{R}^4$.

13. a. $A^{-1} = \begin{bmatrix} -7 & 3 \\ -5 & 2 \end{bmatrix}$ **b.** $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -37 \\ -26 \end{bmatrix}$

15. $x = -7r + 5s + 3t$, $y = 3r - 2s - 2t$, $z = 3r - 2s - t$

17. $\begin{bmatrix} 46 & 33 & 30 \\ 39 & 29 & 26 \\ 99 & 68 & 63 \end{bmatrix}$ **19.** $\begin{bmatrix} 3 & 2 & 1 \\ 0 & 3 & 2 \\ 1 & 3 & 4 \end{bmatrix}$.

21. The matrix is invertible for any value of r except $r = 0$.

23. T T T F T T T F F F

25. a. No; **b.** Yes

27. a. Notice that $A(A^{-1}B) = (AA^{-1})B = IB = B$, so $X = A^{-1}B$ is a solution. To show uniqueness, suppose that $AX = B$. Then $A^{-1}(AX) = A^{-1}B$, $(A^{-1}A)X = A^{-1}B$, $IX = A^{-1}B$, and $X = A^{-1}B$; therefore, this is the only solution.

b. Let $E_1, E_2, \ldots, E_k$ be elementary matrices that reduce $[A \mid B]$ to $[I \mid X]$, and let $C = E_kE_{k-1} \cdots E_2E_1$. Then $CA = I$ and $CB = X$. Thus, $C = A^{-1}$ and so $X = A^{-1}B$.

29. a. $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

39. $\begin{bmatrix} 0.355 & -0.0645 & 0.161 \\ -0.129 & 0.387 & 0.0323 \\ -0.0968 & 0.290 & -0.226 \end{bmatrix}$

41. $\begin{bmatrix} 0.0275 & -0.0296 & -0.0269 & 0.0263 \\ 0.168 & -0.0947 & 0.0462 & -0.0757 \\ 0.395 & -0.138 & -0.00769 & -0.0771 \\ -0.0180 & 0.0947 & 0.00385 & 0.0257 \end{bmatrix}$

43. See answer to Exercise 9.

45. See answer to Exercise 41.

47. $\begin{bmatrix} 0.291 & 0.199 & 0.0419 & -0.00828 & -0.272 \\ -0.0564 & 0.159 & 0.148 & -0.0737 & -0.0699 \\ 0.0276 & 0.145 & -0.00841 & -0.0302 & -0.0250 \\ -0.0467 & 0.122 & -0.029 & 0.133 & -0.0084 \\ 0.0116 & -0.128 & -0.0470 & -0.0417 & 0.178 \end{bmatrix}$

M1. 0.001783 **M3.** 0.4397

M5. −418.07 **M7.** −0.001071

Section 1.6

1. A subspace **3.** Not a subspace

5. Not a subspace **7.** Not a subspace

9. A subspace

13. a. Every subspace of $\mathbb{R}^2$ is either the origin, a line through the origin, or all of $\mathbb{R}^2$.

b. Every subspace of $\mathbb{R}^3$ is either the origin, a line through the origin, a plane through the origin, or all of $\mathbb{R}^3$.

15. No, because $\mathbf{0} = r\mathbf{0}$ for all $r \in \mathbb{R}$, so $\mathbf{0}$ is not a *unique* linear combination of $\mathbf{0}$.

17. $\{[-1, 3, 0], [-1, 0, 3]\}$

19. $\{[-7, 1, 13, 0], [-6, -1, 0, 13]\}$

21. $\{[-60, 137, 33, 0, 1]\}$ **23.** Not a basis

25. A basis **27.** A basis

29. Not a basis **31.** $\{[-1, -3, 11]\}$

33. $\operatorname{sp}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k) = \operatorname{sp}(\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m)$ if and only if each $\mathbf{v}_i$ is a linear combination of the $\mathbf{w}_j$ and each $\mathbf{w}_j$ is a linear combination of the $\mathbf{v}_i$.

35. $\mathbf{x} = \begin{bmatrix} -\frac{5}{8} \\ \frac{7}{4} \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -3r/4 + s/8 \\ 3r/2 + s/4 \\ r \\ s \end{bmatrix}$

37. $\mathbf{x} = \begin{bmatrix} \frac{5}{3} \\ \frac{5}{3} \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} -(5r + s)/3 \\ (r + 2s)/3 \\ r \\ s \end{bmatrix}$

39. In this case, a solution is not uniquely determined. The system is viewed as underdetermined because it is insufficient to determine a unique solution.

41. $x - y = 1$
$2x - 2y = 2$
$3x - 3y = 3$

49. A basis 51. Not a basis

M1. Not a basis M3. A basis

M5. The probability is actually 1.

M7. Yes, we obtained the expected result.

Section 1.7

1. Not a transition matrix

3. Not a transition matrix

5. A regular transition matrix

7. A transition matrix, but not regular

9. 0.28 11. 0.330 13. Not regular

15. Regular 17. Not regular

19. $\begin{bmatrix} \frac{3}{4} \\ \frac{1}{4} \end{bmatrix}$ 21. $\begin{bmatrix} \frac{1}{4} \\ \frac{3}{8} \\ \frac{3}{8} \end{bmatrix}$ 23. $\begin{bmatrix} \frac{12}{35} \\ \frac{8}{35} \\ \frac{15}{35} \end{bmatrix}$

25. T F F T T F T F F T

27. $A^{100} \approx \begin{bmatrix} \frac{12}{35} & \frac{12}{35} & \frac{12}{35} \\ \frac{8}{35} & \frac{8}{35} & \frac{8}{35} \\ \frac{15}{35} & \frac{15}{35} & \frac{15}{35} \end{bmatrix}$ 29. 0.47 31. $\begin{bmatrix} .32 \\ .47 \\ .21 \end{bmatrix}$

33. The Markov chain is regular because T has no zero entry; $\mathbf{s} = \begin{bmatrix} \frac{11}{43} \\ \frac{17}{43} \\ \frac{15}{43} \end{bmatrix}$.

35. $\frac{1}{8}$ 37. $\begin{bmatrix} \frac{1}{4} \\ \frac{1}{2} \\ \frac{1}{4} \end{bmatrix}$

39. The Markov chain is regular because T^2 has no zero entries; $\mathbf{s} = \begin{bmatrix} \frac{1}{4} \\ \frac{1}{2} \\ \frac{1}{4} \end{bmatrix}$.

41. $$T = \begin{array}{c} \begin{array}{ccc} \text{D} & \text{H} & \text{R} \end{array} \\ \begin{bmatrix} 0 & 0 & 0 \\ 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 1 \end{bmatrix} \end{array} \begin{array}{l} \text{D} \\ \text{H}. \\ \text{R} \end{array}$$

The recessive state is absorbing, because there is an entry 1 in the row 3, column 3 position.

45. $\begin{bmatrix} \frac{1}{5} \\ \frac{4}{5} \end{bmatrix}$ 47. $\begin{bmatrix} \frac{5}{9} \\ \frac{4}{9} \end{bmatrix}$ 49. $\begin{bmatrix} .4037 \\ .3975 \\ .1988 \end{bmatrix}$

51. $\begin{bmatrix} .2115 \\ .3462 \\ .4423 \end{bmatrix}$ 53. $\begin{bmatrix} .2682 \\ .2235 \\ .1916 \\ .1676 \\ .1490 \end{bmatrix}$

Section 1.8

1. 0001011 0000000 0111001 1111111
1111111 0100101 0000000 0100101
1111111 0111001

3. $x_4 = x_1 + x_2$, $x_5 = x_1 + x_3$, $x_6 = x_2 + x_3$

5. Note that each x_i appears in at least two parity-check equations and that, for each combination i, j of positions in a message word, some parity-check equation contains one of x_i, x_j but not the other. As explained following Eq. (2) in the text, this shows that the distance between code words is at least 3. Consequently, any two errors can be detected.

7. **a.** 110 **b.** 001 **c.** 110
d. 001 or 100 or 111 **e.** 101

9. We compute, using binary addition,

$$\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix},$$

where the first matrix is the parity-check matrix and the second matrix has as *columns* the received words.

a. Because the first column of the product is the fourth column of the parity-check matrix, we change the fourth position of the received word 110111 to get the code word 110011 and decode it as 110.

b. Because the second column of the product is the zero vector, the received word 001011 is a code word and we decode it as 001.

c. Because the third column of the product is the third column of the parity-check matrix, we change the third position of the received word 111011 to get the code word 110011 and decode it as 110.

d. Because the fourth column of the product is not the zero vector and not a column of the parity-check matrix, there are at least two errors, and we ask for retransmission.

e. Because the fifth column of the product is the third column of the parity-check matrix, we change the third position of the received word 100101 to get the code word 101101 and decode it as 101.

1. Because we add by components, this follows from the fact that, using binary addition, $1 + 1 = 1 - 1 = 0$, $1 + 0 = 1 - 0 = 1$, $0 + 1 = 0 - 1 = 1$, and $0 + 0 = 0 - 0 = 0$.

3. From the solutions of Exercises 11 and 12, we see that $u - v = u + v$ contains 1's in precisely the positions where the two words differ. The number of places where u and v differ is equal to the distance between them, and the number of 1's in $u - v$ is $\mathrm{wt}(u - v)$, so these numbers are equal.

5. This follows immediately from the fact that $\mathbb{B}^n$ itself is closed under addition modulo 2.

7. Suppose that $d(u, v)$ is minimum in C. By Exercise 13, $d(u, v) = \mathrm{wt}(u - v)$, showing that the minimum weight of nonzero code words is less than or equal to the minimum distance between two of them. On the other hand, if w is a nonzero code word of minimum weight, then $\mathrm{wt}(w)$ is the distance from w to the zero code word, showing the opposite inequality, so we have the equality stated in the exercise.

19. The triangle inequality in Exercise 14 shows that if the distance from received word v to code word u is at most m and the distance from v to code word w is at most m, then $d(u, v) \leq 2m$. Thus, if the distance between code words is at least $2m + 1$, a received word v with at most m incorrect components has a *unique* nearest-neighbor code word. This number $2m + 1$ can't be improved, because if $d(u, v) = 2m$, then a possible received word w at distance m from both u and v can be constructed by having its components agree with those of u and v where the components of u and v agree, and by making m of the $2m$ components of w in which u and v differ opposite from the components of u and the other m components of w opposite from those of v.

21. Let e_i be the word in $\mathbb{B}^n$ with 1 in the ith position and 0's elsewhere. Now e_i is not in C, because the distance from e_i to $000 \ldots 0$ is 1, and $000 \ldots 0 \in C$. Also, $v + e_i \neq w + e_j$ for any two distinct words v and w in C, because otherwise $v - w = e_j - e_i$ would be in C with $\mathrm{wt}(e_j - e_i) = 2$. Let $e_i + C = \{e_i + v \mid v \in C\}$. Note that C and $e_i + C$ have the same number of words. The disjoint sets C and $e_i + C$ for $i = 1, 2, \ldots, k$ contain all words whose distances from some word in C are at most 1. Thus $\mathbb{B}^n$ must be large enough to contain all of these $(n + 1)2^k$ elements. That is, $2^n \geq (n + 1)2^k$. Dividing by 2^k gives the desired result.

23. a. 3 **b.** 3 **c.** 4 **d.** 5 **e.** 6 **f.** 7

25. $x_9 = x_1 + x_2 + x_3 + x_6 + x_7$, $x_{10} = x_5 + x_6 + x_7 + x_8$, $x_{11} = x_2 + x_3 + x_4 + x_6 + x_8$, $x_{12} = x_1 + x_3 + x_4 + x_5$ (Other answers are possible.)

CHAPTER 2

Section 2.1

1. Two nonzero vectors in $\mathbb{R}^2$ are dependent if and only if they are parallel.

3. Two nonzero vectors in $\mathbb{R}^3$ are dependent if and only if they are parallel.

5. Three vectors in $\mathbb{R}^3$ are dependent if and only if they all lie in one plane through the origin.

7. $\{[-3, 1], [6, 4]\}$

9. $\{[2, 1], [1, 4]\}$

11. $\{[1, 2, 1, 2], [2, 1, 0, -1]\}$

13. $\{[1, 3, 5, 7], [2, 0, 4, 2]\}$

15. $\{[2, 3, 1], [5, 2, 1]\}$

17. Independent

19. Dependent

21. Independent

23. Dependent

25. Dependent

27. $\{[2, 1, 1, 1], [1, 0, 1, 1], [1, 0, 0, 0], [0, 0, 1, 0]\}$

31. Suppose that $r_1\mathbf{w}_1 + r_2\mathbf{w}_2 + r_3\mathbf{w}_3 = \mathbf{0}$, so that $r_1(2\mathbf{v}_1 + 3\mathbf{v}_2) + r_2(\mathbf{v}_2 - 2\mathbf{v}_3) + r_3(-\mathbf{v}_1 - 3\mathbf{v}_3) = \mathbf{0}$. Then $(2r_1 - r_3)\mathbf{v}_1 + (3r_1 + r_2)\mathbf{v}_2 + (-2r_2 - 3r_3)\mathbf{v}_3 = \mathbf{0}$. We try setting these scalar coefficients to zero and solve the linear system

$$\begin{aligned} 2r_1 \qquad\quad - r_3 &= 0,\\ 3r_1 + \quad r_2 \qquad &= 0,\\ -2r_2 - 3r_3 &= 0. \end{aligned}$$

$$\left[\begin{array}{ccc|c} 2 & 0 & -1 & 0\\ 3 & 1 & 0 & 0\\ 0 & -2 & -3 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -\frac{1}{2} & 0\\ 0 & 1 & \frac{3}{2} & 0\\ 0 & -2 & -3 & 0 \end{array}\right] \sim$$
$$\left[\begin{array}{ccc|c} 1 & 0 & -\frac{1}{2} & 0\\ 0 & 1 & \frac{3}{2} & 0\\ 0 & 0 & 0 & 0 \end{array}\right].$$

This system has the nontrivial solution $r_3 = 2$, $r_2 = -3$, and $r_1 = 1$. Thus, $(2\mathbf{v}_1 + 3\mathbf{v}_2) - 3(\mathbf{v}_2 - 2\mathbf{v}_3) + 2(-\mathbf{v}_1 - 3\mathbf{v}_3) = \mathbf{0}$ for all choices of vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in V$, so $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ are dependent. (Notice that, if the system had only the trivial solution, then any choice of independent vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ would show this exercise to be false.)

33. No such scalars s exist.

35. Let $A = \begin{bmatrix} 1 & 1 & 0\\ 0 & 0 & 0\\ 0 & 0 & 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1\\ 0\\ 0 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0\\ 1\\ 0 \end{bmatrix}$. Then $A\mathbf{v} = A\mathbf{w} = \begin{bmatrix} 1\\ 0\\ 0 \end{bmatrix}$. For the second part of the problem, let $A = \begin{bmatrix} 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 0 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1\\ 0\\ 0 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0\\ 1\\ 0 \end{bmatrix}$. Then $A\mathbf{v} = \mathbf{v}$ and $A\mathbf{w} = \mathbf{w}$, so $A\mathbf{v}$ and $A\mathbf{w}$ are still independent.

39. $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$

41. $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4, \mathbf{u}_6\}$

M1. $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$

M3. $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4, \mathbf{u}_6\}$

Section 2.2

1. a. 2
   b. $\{[2, 0, -3, 1], [3, 4, 2, 2]\}$ by inspection
   c. The set consisting of $\begin{bmatrix} 2\\ 3 \end{bmatrix}, \begin{bmatrix} 0\\ 4 \end{bmatrix}$ or of $\begin{bmatrix} 1\\ 0 \end{bmatrix}, \begin{bmatrix} 0\\ 1 \end{bmatrix}$
   d. The set consisting of $\begin{bmatrix} 12\\ -13\\ 8\\ 0 \end{bmatrix}, \begin{bmatrix} -4\\ -1\\ 0\\ 8 \end{bmatrix}$

3. a. 3
   b. $\{[1, 0, -1, 0], [0, 1, 1, 0], [0, 0, 0, 1]\}$
   c. The set consisting of $\begin{bmatrix} 0\\ 1\\ 4\\ 1 \end{bmatrix}, \begin{bmatrix} 6\\ 2\\ 1\\ 3 \end{bmatrix}, \begin{bmatrix} 3\\ 1\\ 4\\ 0 \end{bmatrix}$
   d. The set consisting of the vector $\begin{bmatrix} 1\\ -1\\ 1\\ 0 \end{bmatrix}$

5. a. 3
   b. $\{[6, 0, 0, 5], [0, 3, 0, 1], [0, 0, 3, 1]\}$
   c. The set consisting of $\begin{bmatrix} 0\\ 2\\ 0 \end{bmatrix}, \begin{bmatrix} 1\\ 1\\ 2 \end{bmatrix}, \begin{bmatrix} 2\\ 0\\ 1 \end{bmatrix}$ or the set of column vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$

. The set consisting of $\begin{bmatrix} -5 \\ -2 \\ -2 \\ 6 \end{bmatrix}$

7. The rank is 3; therefore, the matrix is not invertible.

). The rank is 3; therefore, the matrix is invertible.

I. T F T T F F T F F T

5. No. If A is $m \times n$ and C is $n \times s$, then AC is $m \times s$. The column space of AC lies in $\mathbb{R}^m$ whereas the column space of C lies in $\mathbb{R}^n$.

7. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

9. Let $A = I$ and $C = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. Then rank $(AC) = 1 < 2 =$ rank (A).

5. **a.** 3 **b.** Rows 1, 2, and 4

tion 2.3

I. Yes. We have $T([x_1, x_2, x_3]) = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Example 3 then shows that T is a linear transformation.

3. No. $T(0[1, 2, 3]) = T([0, 0, 0]) = [1, 1, 1, 1]$, whereas $0T([1, 2, 3]) = 0[1, 1, 1, 1] = [0, 0, 0, 0]$.

5. $[24, -34]$

7. $[2, 7, -15]$

9. $[4, 2, 4]$

1. $[802, -477, 398, 57]$

13. $\begin{bmatrix} 1 & 1 \\ 1 & -3 \end{bmatrix}$

5. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

17. $\begin{bmatrix} 1 & -1 & 3 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

9. The matrix associated with $T' \circ T$ is $\begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix}$; $(T' \circ T)([x_1, x_2]) = [2x_1, 3x_1 + x_2]$.

:1. Yes; $T^{-1}([x_1, x_2]) = \left[\frac{3x_1 + x_2}{4}, \frac{x_1 - x_2}{4}\right]$.

23. Yes; $T^{-1}([x_1, x_2, x_3]) = [x_3, x_2 - x_3, x_1 - x_2]$.

25. Yes; $T^{-1}([x_1, x_2, x_3]) = \left[x_3, \frac{-x_1 + 3x_2 - 2x_3}{4}, \frac{x_1 + x_2 - 2x_3}{4}\right]$.

27. Yes; $(T' \circ T)^{-1}([x_1, x_2]) = \left[\frac{x_1}{2}, \frac{-3x_1 + 2x_2}{2}\right]$.

29. T F T T T F T T F T

33. **a.** Because

$$\begin{aligned} T_k(\mathbf{u} + \mathbf{v}) &= [u_1 + v_1, u_2 + v_2, \ldots, \\ &\qquad u_k + v_k, 0, \ldots, 0] \\ &= [u_1, u_2, \ldots, u_k, 0, \ldots, 0] + \\ &\qquad [v_1, v_2, \ldots, v_k, 0, \ldots, 0] \\ &= T_k(\mathbf{u}) + T_k(\mathbf{v}) \end{aligned}$$

and

$$\begin{aligned} T_k(r\mathbf{u}) &= [ru_1, ru_2, \ldots, ru_k, 0, \ldots, 0] \\ &= r[u_1, u_2, \ldots, u_k, 0, \ldots, 0] \\ &= rT_k(\mathbf{u}), \end{aligned}$$

we know that T_k is a linear transformation. We see that $T_k[V] \subseteq W_k$, which means that it is a subspace of W_k, by the box on page 143.

b. Because $T_k[V]$ is a subspace of $T_{k+1}[V]$, we have

$$d_1 \le d_2 \le d_3 \le \cdots \le d_n \le n.$$

If the jth column of H has a pivot but the $(j + 1)$st column has no pivot, then $d_{j+1} = d_j$. However, if the jth and $(j + 1)$st columns both have pivots, then $d_{j+1} = d_j + 1$. See parts c and d for illustrations.

c. If $d_1 = d_2 = 1$ and $d_3 = d_4 = 2$, then H must have the form

$$H = \begin{bmatrix} p & X & X & X \\ 0 & 0 & p & X \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

where a p denotes a pivot and an X denotes a possibly nonzero entry.

d. If $d_1 = 1$, $d_2 = d_3 = d_4 = 2$, and $d_5 = d_6 = 3$, then H must have the form

$$H = \begin{bmatrix} p & X & X & X & X & X \\ 0 & p & X & X & X & X \\ 0 & 0 & 0 & 0 & p & X \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

where a p denotes a pivot and an X denotes a possibly nonzero entry.

e. The number of pivots in H is the number of distinct dimension numbers in the list $d_1, d_2, d_3, \ldots, d_n$. Moreover, a pivot occurs in the $(j+1)$st column if and only if $d_{j+1} = d_j + 1$; the row and column positions of pivots are completely determined by the list $d_1, d_2, d_3, \ldots, d_n$. The pivot in the kth row occurs in the jth column if and only if d_j is the kth *distinct* positive integer in the list $d_1, d_2, d_3, \ldots, d_n$. Because the numbers d_i depend only on A (they are defined just in terms of the row space of A), so do the number and locations of the pivots; the number and locations are the same for all echelon forms of A.

f. Let H be a *reduced* echelon form of A. Part e shows that the number of pivots and their locations depend only on A, and consequently the number of zero rows depends only on A and is the same for all choices of H. Suppose that the pivot in the kth row of H is in column j. Consider now a nonzero row vector in $\mathbb{R}^n$ that has entries zero in all components corresponding to columns of H containing pivots except for the jth component, where the entry is 1; entries in components not corresponding to pivot column locations in H may be arbitrary. We claim that there is a *unique* such vector in the row space of A—namely, the kth row vector of H. Such a vector must be a linear combination of the nonzero rows of H, which span the row space of A. The fact that there are zeros above as well as below pivots in the *reduced* row-echelon form H shows that the only possible such linear combination of nonzero rows of H is 1 times the kth row plus 0 times the other rows. This gives a characterization of the kth nonzero row of H in terms of the row space of A and completes the demonstration that the reduced echelon form of A is unique.

35. [588, 160, 8]

37. Undefined because $T_2 \circ T_3$ is not invertible.

Section 2.4

1. Because $\begin{bmatrix} 1 & -3 \\ 2 & -6 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x - 3y \\ 2x - 6y \end{bmatrix}$, which has the form $\begin{bmatrix} t \\ 2t \end{bmatrix}$, we see that the range of T_A consists of vectors along the line $y = 2x$. Now $T([1, 2]) = [-5, -10] \neq [1, 2]$, and projection onto a line must leave fixed the points that lie on the line, so T_A is not a projection on the line.

3. a. $\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$ b. $\begin{bmatrix} \frac{1}{2} & \sqrt{3}/2 \\ -\sqrt{3}/2 & \frac{1}{2} \end{bmatrix}$

c. $\begin{bmatrix} -\sqrt{3}/2 & \frac{1}{2} \\ -\frac{1}{2} & -\sqrt{3}/2 \end{bmatrix}$

7. $\begin{bmatrix} \dfrac{1 - m^2}{1 + m^2} & \dfrac{2m}{1 + m^2} \\ \dfrac{2m}{1 + m^2} & \dfrac{m^2 - 1}{1 + m^2} \end{bmatrix}$

11. In column-vector notation, we have $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} -y \\ x \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$, which represents a reflection in the line $y = x$ followed by a reflection in the y-axis.

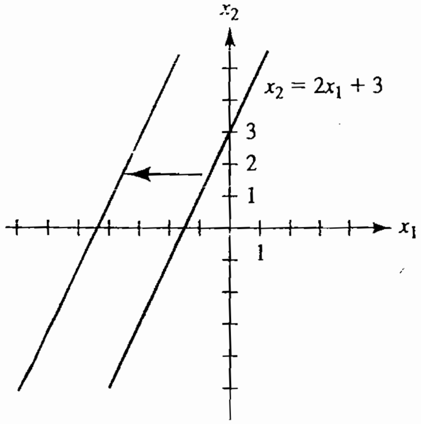
13. In column-vector notation, we have $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} -x \\ -y \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$, which represents a reflection in the x-axis followed by a reflection in the y-axis.

15. In column-vector notation, we have
$$T\left(\begin{bmatrix}x\\y\end{bmatrix}\right) = A\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}1&0\\3&1\end{bmatrix}\begin{bmatrix}1&0\\0&2\end{bmatrix}\begin{bmatrix}1&1\\0&1\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix},$$
which represents a horizontal shear followed by a vertical expansion followed by a vertical shear.

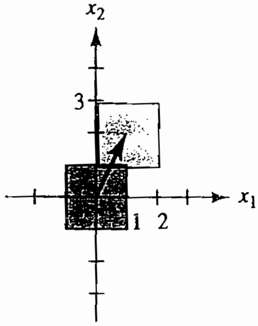
9. $\|\mathbf{v}\| = \sqrt{\mathbf{v}\cdot\mathbf{v}}$ and $\theta = \cos^{-1}\dfrac{\mathbf{u}\cdot\mathbf{v}}{\sqrt{\mathbf{u}\cdot\mathbf{u}}\sqrt{\mathbf{v}\cdot\mathbf{v}}}$.

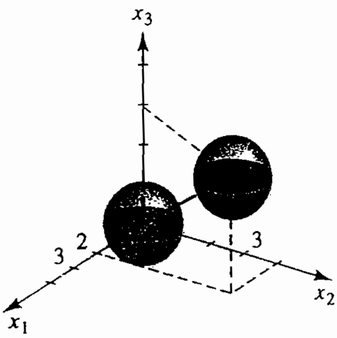
:tion 2.5

1.

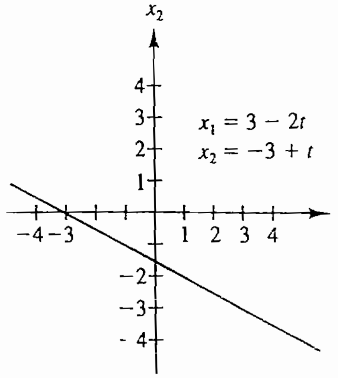


3.





7. $x_1 = 3 - 2t$
$x_2 = -3 + t$



9. $x_1 = \dfrac{d_1c}{d_1^2 + d_2^2} + d_2t,\ x_2 = \dfrac{d_2c}{d_1^2 + d_2^2} - d_1t$

11. a. $x_1 = -2 + t,\ x_2 = 4 - t;$
b. $x_1 = 3 - 3t,\ x_2 = -1 - 2t,\ x_3 = 6 - 7t;$
c. $x_1 = 2 - 3t,\ x_2 = 5t,\ x_3 = 4 - 12t$

13. The lines contain exactly the same set $\{(5 - 3t, -1 + t) \mid t \in \mathbb{R}\}$ of points.

15. a. $\left(\frac{1}{2}, \frac{3}{2}\right)$ b. $\left(\frac{3}{2}, -2, \frac{5}{2}\right)$ c. $\left(-2, \frac{9}{2}, \frac{17}{2}\right)$

17. $\left(-\frac{3}{2}, -\frac{1}{2}, \frac{15}{4}\right)$

19. $\left(\frac{3}{2}, \frac{3}{2}, 1, \frac{7}{2}, -\frac{1}{2}\right)$

21. (2, 3, 4)

23. $x_1 + x_2 + x_3 = 1$

25. $2x_1 + x_2 - x_3 = 2$

27. $x_1 + 4x_2 + x_3 = 10$

29. $x_2 + x_3 = 3,\ -3x_1 - 7x_2 + 8x_3 + 3x_4 = 0$ (There are infinitely many other correct linear systems.)

31. $9x_1 - x_2 + 2x_3 - 6x_4 + 3x_5 = 6$

33. $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 1$

35. The 0-flat $\mathbf{x} = \begin{bmatrix}-1\\0\\2\end{bmatrix}$

37. The 1-flat $\mathbf{x} = \begin{bmatrix}-1\\-1\\0\end{bmatrix} + t\begin{bmatrix}2\\1\\1\end{bmatrix}$

39. The 1-flat $\mathbf{x} = \begin{bmatrix}-8\\-23\\-7\\0\end{bmatrix} + t\begin{bmatrix}0\\-5\\1\\2\end{bmatrix}$

41. The 0-flat $\mathbf{x} = \begin{bmatrix} -43 \\ -12 \\ 7 \\ 1 \end{bmatrix}$

43. T F F T F T F T F T

CHAPTER 3

Section 3.1

1. Not a vector space 3. A vector space

5. Not a vector space 7. Not a vector space

9. A vector space 11. A vector space

13. A vector space 15. A vector space

17. a. $[-1, 0]$ is the "zero vector."
    b. Part 5 of Theorem 3.1 in this vector space becomes $r[-1, 0] = [-1, 0]$, for all $r \in \mathbb{R}$. That is, $[0, 0]$ is not the zero vector $\mathbf{0}$ in this vector space.

27. Both 2×6 matrices and 3×4 matrices contain 12 entries. If we number entries in some fashion from 1 to 12, say starting at the upper left-hand corner and proceeding down each column in turn, then each matrix can be viewed as defining a function from the set $S = \{1, 2, \ldots, 12\}$ into the set $\mathbb{R}$. The rules for adding matrices and multiplying by a scalar correspond in each case to our definitions of addition and scalar multiplication in the space of functions mapping S into $\mathbb{R}$. Thus we can identify both $M_{2,6}$ and $M_{3,4}$ with this function space, and hence with each other.

29. $\mathbb{R}^{24}$ $\mathbb{R}^{25}$ $\mathbb{R}^{26}$ P_{26} $M_{4,7}$
    $M_{2,12}$ P_{24} P_{25} $M_{3,9}$
    $M_{4,6}$ $M_{5,5}$ $M_{2,13}$
    $M_{3,8}$

Section 3.2

1. Not a subspace 3. Not a subspace

5. A subspace

7. a. Because $1 = \sin^2 x + \cos^2 x$, we have $c = c(\sin^2 x) + c(\cos^2 x)$, which shows that $c \in \mathrm{sp}(\sin^2 x, \cos^2 x)$.
   b. Now $\cos 2x = \cos^2 x - \sin^2 x = (-1)\sin^2 x + (1)\cos^2 x$, which shows that $\cos 2x \in \mathrm{sp}(\sin^2 x, \cos^2 x)$
   c. Now $\cos 4x = \cos^2 2x - \sin^2 2x = (1 - \sin^2 2x) - \sin^2 2x = \frac{1}{7}(7) + (-2)\sin^2 2x$, which shows that $\cos 4x$, and thus $8 \cos 4x$, is in $\mathrm{sp}(7, \sin^2 2x)$.

9. a. We see that $\mathbf{v}_1, 2\mathbf{v}_1 + \mathbf{v}_2 \in \mathrm{sp}(\mathbf{v}_1, \mathbf{v}_2)$; and therefore,
   $$\mathrm{sp}(\mathbf{v}_1, 2\mathbf{v}_1 + \mathbf{v}_2) \subseteq \mathrm{sp}(\mathbf{v}_1, \mathbf{v}_2).$$
   Furthermore, $\mathbf{v}_1 = 1\mathbf{v}_1 + 0(2\mathbf{v}_1 + \mathbf{v}_2)$ and $\mathbf{v}_2 = (-2)\mathbf{v}_1 + 1(2\mathbf{v}_1 + \mathbf{v}_2)$, showing that $\mathbf{v}_1, \mathbf{v}_2 \in \mathrm{sp}(\mathbf{v}_1, 2\mathbf{v}_1 + \mathbf{v}_2)$; and therefore,
   $$\mathrm{sp}(\mathbf{v}_1, \mathbf{v}_2) \subseteq \mathrm{sp}(\mathbf{v}_1, 2\mathbf{v}_1 + \mathbf{v}_2).$$
   Thus, $\mathrm{sp}(\mathbf{v}_1, \mathbf{v}_2) = \mathrm{sp}(\mathbf{v}_1, 2\mathbf{v}_1 + \mathbf{v}_2)$.

11. Dependent 13. Dependent

15. Independent 17. Dependent

19. Independent 21. Not a basis

23. $\{1, 4x + 3, x^2 + 2\}$

25. T F T T T F F F T T

35. Let $W = \mathrm{sp}(\mathbf{e}_1, \mathbf{e}_2)$ and $U = \mathrm{sp}(\mathbf{e}_3, \mathbf{e}_4, \mathbf{e}_5)$ in $\mathbb{R}^5$. Then $W \cap U = \{\mathbf{0}\}$ and each $\mathbf{x} \in \mathbb{R}^5$ has the form $\mathbf{x} = \mathbf{w} + \mathbf{u}$, where $\mathbf{w} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2$ and $\mathbf{u} = x_3\mathbf{e}_3 + x_4\mathbf{e}_4 + x_5\mathbf{e}_5$.

39. In deciding whether $\sin x$ and $\cos x$ are independent, we consider linear combinations of them *with scalar coefficients*. The given coefficients $f(x)$ and $g(x)$ are not scalars. For a counterexample, consider $f(x) = -\cos x$ and $g(x) = \sin x$. We have $(-\cos x)(\sin x) + (\sin x)(\cos x) = 0$.

41. The set of solutions consists of all functions of the form $h(x) + p(x)$, where $h(x)$ is the general solution of the corresponding homogeneous equation $f_n(x)y^{(n)} + f_{n-1}(x)y^{(n-1)} + \cdots + f_2(x)y'' + f_1(x)y' + f_0(x)y = 0$.

43. a. $\{a \sin x + b \cos x \mid a, b \in \mathbb{R}\}$
    b. $\{a \sin x + b \cos x + x \mid a, b \in \mathbb{R}\}$

45. a. $\{ae^{3x} + be^{-3x} + c \mid a, b, c \in \mathbb{R}\}$
    b. $\{ae^{3x} + be^{-3x} + c - \frac{x^3}{27} - \frac{1}{9}x^2 - \frac{2}{81}x \mid a, b, c \in \mathbb{R}\}$

7. One basis B for W consists of those f_a for a in S defined by $f_a(a) = 1$ and $f_a(s) = 0$ for $s \neq a$ in S. If $f \in W$ and $f(s) \neq 0$ only for $s \in \{a_1, a_2, \ldots, a_n\}$, then $f = f(a_1)f_{a_1} + f(a_2)f_{a_2} + \cdots + f(a_n)f_{a_n}$. Now the linear combination $g = c_1f_{b_1} + c_2f_{b_2} + \cdots + c_mf_{b_m}$ is a function satisfying $g(b_j) = c_j$ for $j = 1, 2, \ldots, m$ and $g(s) = 0$ for all other $s \in S$. Thus $g \in W$, so B spans only W. (The crucial thing is that all linear combinations are sums of only *finite* numbers of vectors.)

tion 3.3

1. $[1, -1]$ 3. $[2, 6, -4]$ 5. $[2, 1, 3]$

7. $[4, 1, -2, 1]$ 9. $\left[\frac{1}{2}, \frac{1}{2}, 1, -\frac{1}{2}, 0\right]$

11. $[1, 2, -1, 5]$

13.
$$\begin{aligned} p(x) &= [(x+1)-1]^3 + [(x+1)-1]^2 - [(x+1)-1] - 1 \\ &= (x+1)^3 - 3(x+1)^2 + 3(x+1) - 1 \\ &\quad + (x+1)^2 - 2(x+1) + 1 \\ &\quad - (x+1) + 1 \\ &\quad - 1 \\ &= (x+1)^3 - 2(x+1)^2 + 0(x+1) + 0 \end{aligned}$$

so the coordinate vector is $[1, -2, 0, 0]$.

15. $[4, 3, -5, -4]$

17. Let $x^3 - 4x^2 + 3x + 7 = b_3(x-2)^3 + b_2(x-2)^2 + b_1(x-2) + b_0$. Setting $x = 2$, we find that $8 - 16 + 6 + 7 = b_0$, so $b_0 = 5$. Differentiating and setting $x = 2$, we obtain $3x^2 - 8x + 3 = 3b_3(x-2)^2 + 2b_2(x-2) + b_1$; $12 - 16 + 3 = b_1$; $b_1 = -1$. Differentiating again and setting $x = 2$, we have $6x - 8 = 6b_3(x-2) + 2b_2$; $12 - 8 = 2b_2$; $b_2 = 2$. Differentiating again, we obtain $6 = 6b_3$, so $b_3 = 1$. The coordinate vector is $[1, 2, -1, 5]$.

19. **b.** $\{f_1(x), f_2(x)\}$ 21. $2x^2 + 6x + 2$

ction 3.4

1. A linear transformation, $\ker(T) = \{f \in F \mid f(-4) = 0\}$, not invertible

3. A linear transformation, $\ker(T)$ is the zero function, invertible

5. A linear transformation, $\ker(T)$ is the zero function, invertible

7. The zero function is the only function in $\ker(T)$.

9. $\{c_1e^{2x} + c_2e^{-2x} - \frac{1}{5}\sin x \mid c_1, c_2 \in \mathbb{R}\}$

11. $\{c_1x + c_2 + \cos x \mid c_1, c_2 \in \mathbb{R}\}$

13. $\{c_1 \sin 2x + c_2 \cos 2x + \frac{1}{4}x^2 - \frac{1}{8} \mid c_1, c_2 \in \mathbb{R}\}$

15. $\{c_1e^{2x} + c_2x + c_3 - \frac{1}{12}x^3 - \frac{1}{8}x^2 \mid c_1, c_2, c_3 \in \mathbb{R}\}$

17. $T(\mathbf{v}) = 4\mathbf{b}_1' + 4\mathbf{b}_2' + 7\mathbf{b}_3' + 6\mathbf{b}_4'$

19. $T(\mathbf{v}) = -17\mathbf{b}_1' + 4\mathbf{b}_2' + 13\mathbf{b}_3' - 3\mathbf{b}_4'$

21. **a.** $A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

b. $A\begin{bmatrix} 4 \\ -5 \\ 10 \\ -13 \end{bmatrix} = \begin{bmatrix} 0 \\ 12 \\ -10 \\ 10 \end{bmatrix}$; $12x^2 - 10x + 10$

c. The second derivative is given by $A^2\begin{bmatrix} -5 \\ 8 \\ -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -30 \\ 16 \end{bmatrix}$; $-30x + 16$

23. **a.** $D(x^2e^x) = x^2e^x + 2xe^x$; $D^2(x^2e^x) = x^2e^x + 4xe^x + 2e^x$; $D(xe^x) = xe^x + e^x$; $D^2(xe^x) = xe^x + 2e^x$; $D(e^x) = e^x$; $D^2(e^x) = e^x$

$A = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 2 & 2 & 1 \end{bmatrix}$

b. From the computations in part a, we have $A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, and computation shows that $A_1^2 = A$.

25. We obtain $A = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$ in both part a and part b.

27. $\begin{bmatrix} 9 & 0 & 0 \\ 0 & 25 & 0 \\ 0 & 0 & 81 \end{bmatrix}$

29. $(a + c)e^{2x} + be^{4x} + (a + c)e^{8x}$ 31. $\begin{bmatrix} -3 & -3 \\ 4 & -4 \end{bmatrix}$

33. $-2b \sin 2x + 2a \cos 2x$

45. If A_1 is the representation of T_1 and A_2 is the representation of T_2 relative to B,B', then $A_1 + A_2$ is the representation of $T_1 + T_2$ relative to B,B'. If A is the representation of T relative to B,B', then rA is the representation of rT relative to B,B'

51. Let $V = D_\infty$, the space of infinitely differentiable functions mapping $\mathbb{R}$ into $\mathbb{R}$. Let $T(f) = f'$ for $f \in D_\infty$. Then range$(T) = D_\infty$ because every infinitely differentiable function is continuous and thus has an antiderivative, but $T(x) = T(x + 1)$ shows that T is not one-to-one.

Section 3.5

1. Not an inner product
3. Not an inner product
5. Not an inner product
7. An inner product
9. Not an inner product

11. a. $\frac{5}{6}$ b. $\frac{1}{\sqrt{3}}$ c. $\frac{1}{\sqrt{30}}$ d. $\frac{1}{\sqrt{2}}$

13. $f(x) = -x + \frac{1}{2}$ and $g(x) = \cos \pi x$. Other answers are possible.

15. 77

CHAPTER 4

Section 4.1

1. -15 3. 15

5. $$\mathbf{b} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} b_1 & b_2 & b_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = b_1(b_2c_3 - b_3c_2) - b_2(b_1c_3 - b_3c_1) + b_3(b_1c_2 - b_2c_1) = 0,$$

$$\mathbf{c} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = c_1(b_2c_3 - b_3c_2) - c_2(b_1c_3 - b_3c_1) + c_3(b_1c_2 - b_2c_1) = 0$$

7. 120 9. -9

11. $$\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1 = -(b_1a_2 - b_2a_1) = -\begin{vmatrix} b_1 & b_2 \\ a_1 & a_2 \end{vmatrix}$$

13. $-6\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$ 15. $0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}$

17. $22\mathbf{i} + 18\mathbf{j} + 2\mathbf{k}$

19. F T T F F T F T T F

21. 38 23. $\sqrt{62}$ 25. $\frac{19}{2}$

27. $\frac{\sqrt{230}}{2}$ 29. 16 31. $\sqrt{390}$

33. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = -6$,
$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = 12\mathbf{i} + 4\mathbf{k}$

35. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 19$,
$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = 3\mathbf{i} - 7\mathbf{j} + \mathbf{k}$

37. 20 39. 9 41. 1 43. $\frac{7}{3}$

45. Not collinear 47. Collinear

49. Not coplanar 51. Not coplanar

53. 0 55. $\|\mathbf{a}\|^2\|\mathbf{b}\|^2$

57. $\mathbf{i} \times (\mathbf{i} \times \mathbf{j}) = \mathbf{i} \times \mathbf{k} = -\mathbf{j}$, but $(\mathbf{i} \times \mathbf{i}) \times \mathbf{j} = \mathbf{0} \times \mathbf{j} = \mathbf{0}$.

59. $$\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = (b_2c_3 - b_3c_2)\mathbf{i} - (b_1c_3 - b_3c_1)\mathbf{j} + (b_1c_2 - b_2c_1)\mathbf{k}.$$ Thus,
$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$
Equation (4) in the text shows that this determinant is $\pm$(Volume of the box determined by **a**, **b**, and **c**). Similarly,
$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = \begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = c_1(a_2b_3 - a_3b_2) - c_2(a_1b_3 - a_3b_1) + c_3(a_1b_2 - a_2b_1),$$
which is the same number.

61. 322 **63.** 130.8097

M3. **a.** $67.38\mathbf{i} - 88.76\mathbf{j} - 8.11\mathbf{k}$

b. 111.7326

Section 4.2

1. 13 **3.** -21 **5.** 19 **7.** 320

9. 0 **11.** -6 **13.** 2 **15.** 4

17. 54 **19.** $\frac{1}{2}$

21. F T T F T T T F T F T

23. 0 **25.** 6 **27.** 5, -2 **29.** 1, 6, -2

33. Let A have column vectors $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$. Let

$$B = \begin{bmatrix} b_1 & & & \\ & b_2 & & \mathbf{0} \\ & & \ddots & \\ \mathbf{0} & & & b_n \end{bmatrix}.$$

Then $AB = \begin{bmatrix} | & | & & | \\ b_1\mathbf{a}_1 & b_2\mathbf{a}_2 & \cdots & b_n\mathbf{a}_n \\ | & | & & | \end{bmatrix}$.

Applying the column scaling property n times, we have

$$\det(AB) = b_1 \begin{vmatrix} | & | & & | \\ \mathbf{a}_1 & b_2\mathbf{a}_2 & \cdots & b_n\mathbf{a}_n \\ | & | & & | \end{vmatrix}$$

$$= b_1b_2 \begin{vmatrix} | & | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & b_3\mathbf{a}_3 & \cdots & b_n\mathbf{a}_n \\ | & | & | & & | \end{vmatrix}$$

$$= \cdots = (b_1b_2 \cdots b_n)\det(A)$$

$$= \det(B)\det(A).$$

Section 4.3

1. 27 **3.** -8 **5.** -195

7. -207 **9.** 496

13. Let $R_1, R_2, \ldots, R_k$ be square submatrices positioned corner-to-corner along the diagonal of a matrix A, where A has zero entries except possibly for those in the submatrices R_i. Then $\det(A) = \det(R_1)\det(R_2)\cdots\det(R_k)$.

15. $A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -1 & 2 \end{bmatrix}$

17. $A^{-1} = -\frac{1}{17}\begin{bmatrix} 1 & 4 & -4 \\ 7 & -6 & -11 \\ -5 & -3 & 3 \end{bmatrix}$

19. $A^{-1} = -\frac{1}{13}\begin{bmatrix} -7 & 5 & 1 \\ 4 & -1 & -8 \\ -1 & -3 & 2 \end{bmatrix}$

21. $\operatorname{adj}(A) = \begin{bmatrix} -7 & -1 & 4 \\ -3 & 2 & -8 \\ 6 & -4 & -1 \end{bmatrix}$

25. The coefficient matrix has determinant 0 so the system cannot be solved using Cramer's rule.

27. $x_1 = 0, x_2 = 1$

29. $x_1 = 3, x_2 = -\frac{11}{3}, x_3 = -\frac{7}{3}$

31. $x_1 = 0, x_2 = \frac{1}{2}, x_3 = 0$

33. $x_2 = \frac{41}{59}$

35. F T T F T T T F T T F

41. 19095. The answers are the same.

43. With default roundoff, the smallest such value of m is 6, which gave 0 rather than 1 as the value of the determinant. Although this wrong result would lead us to believe that A^6 is singular, the error is actually quite small compared with the size of the entries in A^6. We obtained 0 as $\det(A^m)$ for $6 \le m \le 17$, and we obtained -1.470838×10^{10} as $\det(A^{18})$. MATCOMP gave us -1.598896×10^{13} as $\det(A^{20})$.

With roundoff control 0, we obtained $\det(A^6) = 1$ and $\det(A^7) = 0.999998$, which has an error of only 0.000002. The error became significantly larger with $m = 12$, where we obtained 11.45003 for the determinant. We obtained the same result with $m = 20$ as we did with the default roundoff.

With the default roundoff, computed entries of magnitude less than (0.0001)(Smallest nonzero magnitude in A^m) are set equal to zero when A^m is reduced to echelon form. As soon as m is large enough (so that the entries of A^m are large enough), this creates a false zero entry on the diagonal, which produces a false calculation of 0 for the determinant. With roundoff zero, this does not happen. But when m get sufficiently large, roundoff error in the computation of A^m and in its reduction to echelon form creates even greater error in the calculation of the determinant, no matter what roundoff control number is taken. Notice, however, that the value given for the determinant is always small compared with the size of the entries in A^m.

45. $\begin{bmatrix} 12 & -8 & -7 \\ -11 & 13 & 5 \\ 9 & -6 & -1 \end{bmatrix}$

47. $\begin{bmatrix} -4827 & 114 & -2211 & 2409 \\ -3218 & 76 & -1474 & 1606 \\ 8045 & -190 & 3685 & -4015 \\ 1609 & -38 & 737 & -803 \end{bmatrix}$

Section 4.4

1. $\sqrt{213}$ **3.** $\sqrt{30}$ **5.** 11 **7.** 38

9. 2 **11.** $\frac{\sqrt{6}}{2}$ **13.** $\frac{2}{3}$

15. Let **a**, **b**, **c**, and **d** be the vectors from the origin to the four points. Let A be the $n \times 3$ matrix with column vectors $\mathbf{b} - \mathbf{a}$, $\mathbf{c} - \mathbf{a}$, and $\mathbf{d} - \mathbf{a}$. The points are coplanar if and only if $\det(A^TA) = 0$.

17. The points are not coplanar. **19.** 32

21. 144π **23.** 264 **25.** $\frac{2816\pi}{3}$

27. $5\sqrt{3}$ **29.** $25\pi\sqrt{3}$ **31.** $16\sqrt{17}$

33. **a.** 0 **b.** 0

35. T F T F T F T T F T

CHAPTER 5

Section 5.1

1. $\mathbf{v}_1$, $\mathbf{v}_3$, and $\mathbf{v}_5$ are eigenvectors of A_1, with corresponding eigenvalues -1, 2, and 2, respectively. $\mathbf{v}_2$, $\mathbf{v}_3$, and $\mathbf{v}_4$ are eigenvectors of A_2, with corresponding eigenvalues 5, 5, and 1, respectively.

3. *Characteristic polynomial*: $\lambda^2 - 4\lambda + 3$
Eigenvalues: $\lambda_1 = 1, \lambda_2 = 3$
Eigenvectors: for $\lambda_1 = 1$: $\mathbf{v}_1 = \begin{bmatrix} -r \\ r \end{bmatrix}, r \neq 0,$
for $\lambda_2 = 3$: $\mathbf{v}_2 = \begin{bmatrix} -s \\ 2s \end{bmatrix}, s \neq 0$

5. *Characteristic polynomial*: $\lambda^2 + 1$
Eigenvalues: There are no real eigenvalues.

7. *Characteristic polynomial*: $-\lambda^3 + 2\lambda^2 + \lambda - 2$
Eigenvalues: $\lambda_1 = -1, \lambda_2 = 1, \lambda_3 = 2$
Eigenvectors: for $\lambda_1 = -1$: $\mathbf{v}_1 = \begin{bmatrix} 0 \\ r \\ 0 \end{bmatrix}, r \neq 0,$
for $\lambda_2 = 1$: $\mathbf{v}_2 = \begin{bmatrix} 0 \\ -s \\ s \end{bmatrix}, s \neq 0,$
for $\lambda_3 = 2$: $\mathbf{v}_3 = \begin{bmatrix} -t \\ -t \\ t \end{bmatrix}, t \neq 0$

9. *Characteristic polynomial*: $-\lambda^3 + 8\lambda^2 + \lambda - 8$
Eigenvalues: $\lambda_1 = -1, \lambda_2 = 1, \lambda_3 = 8$
Eigenvectors: for $\lambda_1 = -1$: $\mathbf{v}_1 = \begin{bmatrix} 0 \\ r \\ 0 \end{bmatrix}, r \neq 0,$
for $\lambda_2 = 1$: $\mathbf{v}_2 = \begin{bmatrix} 0 \\ -s \\ s \end{bmatrix}, s \neq 0$
for $\lambda_3 = 8$: $\mathbf{v}_3 = \begin{bmatrix} -t \\ -t \\ t \end{bmatrix}, t \neq 0$

11. *Characteristic polynomial*: $-\lambda^3 - \lambda^2 + 8\lambda + 12$
Eigenvalues: $\lambda_1 = \lambda_2 -2, \lambda_3 = 3$
Eigenvectors: for $\lambda_1, \lambda_2 = -2$: $\mathbf{v}_1 = \begin{bmatrix} -r \\ s \\ r \end{bmatrix},$

r and s not both 0,

for $\lambda_3 = 3$: $\mathbf{v}_3 = \begin{bmatrix} 0 \\ -t \\ t \end{bmatrix}$, $t \neq 0$

13. *Characteristic polynomial*: $-\lambda^3 + 2\lambda^2 + 4\lambda - 8$
Eigenvalues: $\lambda_1 = -2$, $\lambda_2 = \lambda_3 = 2$
Eigenvectors: for $\lambda_1 = -2$: $\mathbf{v}_1 = \begin{bmatrix} -r \\ -3r \\ r \end{bmatrix}$, $r \neq 0$,
for $\lambda_2, \lambda_3 = 2$: $\mathbf{v}_2 = \begin{bmatrix} 0 \\ s \\ 0 \end{bmatrix}$, $s \neq 0$

15. *Characteristic polynomial*: $-\lambda^3 - 3\lambda^2 + 4$
Eigenvalues: $\lambda_1 = \lambda_2 = -2$, $\lambda_3 = 1$
Eigenvectors: for $\lambda_1, \lambda_2 = -2$: $\mathbf{v}_1 = \begin{bmatrix} 0 \\ r \\ 0 \end{bmatrix}$, $r \neq 0$,
for $\lambda_3 = 1$: $\mathbf{v}_3 = \begin{bmatrix} 3s \\ -s \\ 3s \end{bmatrix}$, $s \neq 0$

17. *Eigenvalues*: $\lambda_1 = -1$, $\lambda_2 = 5$
Eigenvectors: for $\lambda_1 = -1$: $\mathbf{v}_1 = \begin{bmatrix} r \\ r \end{bmatrix}$, $r \neq 0$,
for $\lambda_2 = 5$: $\mathbf{v}_2 = \begin{bmatrix} -s \\ s \end{bmatrix}$, $s \neq 0$

19. *Eigenvalues*: $\lambda_1 = 0$, $\lambda_2 = 1$, $\lambda_3 = 2$
Eigenvectors: for $\lambda_1 = 0$: $\mathbf{v}_1 = \begin{bmatrix} -r \\ 0 \\ r \end{bmatrix}$, $r \neq 0$,
for $\lambda_2 = 1$: $\mathbf{v}_2 = \begin{bmatrix} 0 \\ s \\ 0 \end{bmatrix}$, $s \neq 0$,
for $\lambda_3 = 2$: $\mathbf{v}_3 = \begin{bmatrix} t \\ 0 \\ t \end{bmatrix}$, $t \neq 0$

21. *Eigenvalues*: $\lambda_1 = -2$, $\lambda_2 = 1$, $\lambda_3 = 5$
Eigenvectors: for $\lambda_1 = -2$: $\mathbf{v}_1 = \begin{bmatrix} 0 \\ r \\ r \end{bmatrix}$, $r \neq 0$,
for $\lambda_2 = 1$: $\mathbf{v}_2 = \begin{bmatrix} -6s \\ 5s \\ 8s \end{bmatrix}$, $s \neq 0$,
for $\lambda_3 = 5$: $\mathbf{v}_3 = \begin{bmatrix} 0 \\ -5t \\ 2t \end{bmatrix}$, $t \neq 0$

23. F F T T F T F T F T

31. *Eigenvalues*: $\lambda_1 = \dfrac{1 + \sqrt{5}}{2}$, $\lambda_2 = \dfrac{1 - \sqrt{5}}{2}$
Eigenvectors: for $\lambda_1 = \dfrac{1 + \sqrt{5}}{2}$:
$\mathbf{v}_1 = r\begin{bmatrix} (1 + \sqrt{5})/2 \\ 1 \end{bmatrix}$, $r \neq 0$,
for $\lambda_2 = \dfrac{1 - \sqrt{5}}{2}$:
$\mathbf{v}_2 = s\begin{bmatrix} (1 - \sqrt{5})/2 \\ 1 \end{bmatrix}$, $s \neq 0$

33. a. Work with the matrix $A + 10I$, whose eigenvalues are approximately 30, 12, 7, and -9.5. (Other answers are possible.);
b. Work with the matrix $A - 10I$, whose eigenvalues are approximately -9.5, -8, -13, and -30. (Other answers are possible.)

37. When the determinant of an $n \times n$ matrix A is expanded according to the definition using expansion by minors, a sum of $n!$ (signed) terms is obtained, each containing a product of one element from each row and from each column. (This is readily proved by induction on n.) One of the $n!$ terms obtained by expanding $|A - \lambda I|$ to obtain $p(\lambda)$ is

$(a_{11} - \lambda)(a_{22} - \lambda)(a_{33} - \lambda) \cdots (a_{nn} - \lambda)$.

We claim that this is the only one of the $n!$ terms that can contribute to the coefficient of λ^{n-1} in $p(\lambda)$. Any term contributing to the coefficient of λ^{n-1} must contain at least $n - 1$ of the factors in the preceding product; the other factor must be from the remaining row and column, and hence it must be the remaining factor from the diagonal of $A - \lambda I$. Computing the coefficient of λ^{n-1} in the preceding product, we find that, when $-\lambda$ is chosen from all but the factor $a_{ii} - \lambda$ in expanding the product, the resulting contribution to the coefficient of λ^{n-1} in $p(\lambda)$ is $(-1)^{n-1}a_{ii}$. Thus, the coefficient of λ^{n-1} in $p(\lambda)$ is

$(-1)^{n-1}(a_{11} + a_{22} + a_{33} + \cdots + a_{nn})$.

Now $p(\lambda) = (-1)^n(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3) \cdots (\lambda - \lambda_n)$, and computation of this product shows in the same way that the coefficient of λ^{n-1} is also

$(-1)^{n-1}(\lambda_1 + \lambda_2 + \lambda_3 + \cdots + \lambda_n)$.

Thus, $\operatorname{tr}(A) = \lambda_1 + \lambda_2 + \lambda_3 + \cdots + \lambda_n$.

39. Because $\begin{vmatrix} 2-\lambda & -1 \\ 1 & 3-\lambda \end{vmatrix} = \lambda^2 - 5\lambda + 7$, we compute

$$A^2 - 5A + 7I = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}^2 - 5\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} + 7\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -5 \\ 5 & 8 \end{bmatrix} + \begin{bmatrix} -10 & 5 \\ -5 & -15 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

illustrating the Cayley–Hamilton theorem.

43. The desired result is stated as Theorem 5.3 in Section 5.2. Because an $n \times n$ matrix is the standard matrix representation of the linear transformation $T: \mathbb{R}^n \to \mathbb{R}^n$ and the eigenvalues and eigenvectors of the transformation are the same as those of the matrix, proof of Exercise 42 for transformations implies the result for matrices, and vice versa.

45. **a.** $F_8 = 21$;

b. $F_{30} = 832040$;

c. $F_{50} = 12586269025$;

d. $F_{77} = 5527939700884757$;

e. $F_{150} \approx 9.969216677189304 \times 10^{30}$

47. **a.** 0, 1, 2, 1, −3, −7, −4, 10, 25;

b. $\begin{bmatrix} 2 & -3 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$;

c. $a_{30} = -191694$

49. $\lambda_1 = -1$, $\mathbf{v}_1 = \begin{bmatrix} r \\ -r \\ r \\ r \end{bmatrix}$, $r \neq 0$; $\lambda_2 = 3$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ s \end{bmatrix}$, $s \neq 0$; $\lambda_3 = \lambda_4 = 2$, $\mathbf{v}_3 = \begin{bmatrix} 0 \\ -t \\ u \\ t \end{bmatrix}$, t and u not both zero

51. $\lambda_1 = 16$, $\mathbf{v}_1 = \begin{bmatrix} 0 \\ r \\ 0 \\ 0 \end{bmatrix}$, $r \neq 0$; $\lambda_2 = 36$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ s \\ s \end{bmatrix}$, $s \neq 0$; $\lambda_3 = \lambda_4 = 4$, $\mathbf{v}_3 = \begin{bmatrix} t+u \\ t \\ t \\ u \end{bmatrix}$, t and u not both zero

53. *Characteristic polynomial*: $-\lambda^3 - 16\lambda^2 + 48\lambda - 261$

Eigenvalues: $\lambda_1 \approx 1.6033 + 3.3194i$, $\lambda_2 \approx 1.6033 - 3.3194i$, $\lambda_3 \approx -19.2067$

55. *Characteristic polynomial*: $\lambda^4 - 56\lambda^3 + 210\lambda^2 + 22879\lambda + 678658$

Eigenvalues: $\lambda_1 \approx -12.8959 + 13.3087i$,
$\lambda_2 \approx -12.8959 - 13.3087i$,
$\lambda_3 \approx 40.8959 + 17.4259i$,
$\lambda_4 \approx 40.8959 - 17.4259i$

M1. **a.** $\lambda^3 + 16\lambda^2 - 48\lambda + 261$;
$\lambda_1 \approx -19.2067$,
$\lambda_2 \approx 1.6033 + 3.3194i$,
$\lambda_3 \approx 1.6033 - 3.3194i$

b. $\lambda^4 + 14\lambda^3 - 131\lambda^2 + 739\lambda - 21533$;
$\lambda_1 \approx -22.9142$,
$\lambda_2 \approx 10.4799$,
$\lambda_3 \approx -0.7828 + 9.4370i$,
$\lambda_4 \approx -0.7828 - 9.4370i$

M3. λ of minimum magnitude $\approx -2.68877026277667$

M7. λ of maximum magnitude ≈ 19.2077
λ of minimum magnitude ≈ 8.0147

Section 5.2

1. *Eigenvalues*: $\lambda_1 = -5$, $\lambda_2 = 5$

Eigenvectors: for $\lambda_1 = -5$: $\mathbf{v}_1 = \begin{bmatrix} -2r \\ r \end{bmatrix}$, r

for $\lambda_2 = 5$: $\mathbf{v}_2 = \begin{bmatrix} s \\ 2s \end{bmatrix}$, $s \neq 0$

$C = \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}$, $D = \begin{bmatrix} -5 & 0 \\ 0 & 5 \end{bmatrix}$

3. *Eigenvalues*: $\lambda_1 = -1, \lambda_2 = 3$

Eigenvectors: for $\lambda_1 = -1$: $\mathbf{v}_1 = \begin{bmatrix} -r \\ r \end{bmatrix}, r \neq 0,$

for $\lambda_2 = 3$: $\mathbf{v}_2 = \begin{bmatrix} -2s \\ s \end{bmatrix}, s \neq 0$

$$C = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}, D = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$$

5. *Eigenvalues*: $\lambda_1 = -3, \lambda_2 = 1, \lambda_3 = 7$

Eigenvectors: for $\lambda_1 = -3$: $\mathbf{v}_1 = \begin{bmatrix} r \\ 0 \\ 0 \end{bmatrix}, r \neq 0,$

for $\lambda_2 = 1$: $\mathbf{v}_2 = \begin{bmatrix} s \\ s \\ s \end{bmatrix}, s \neq 0,$

for $\lambda_3 = 7$: $\mathbf{v}_3 = \begin{bmatrix} t \\ t \\ 0 \end{bmatrix}, t \neq 0$

$$C = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}, D = \begin{bmatrix} -3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

7. *Eigenvalues*: $\lambda_1 = -1, \lambda_2 = 1, \lambda_3 = 2$

Eigenvectors: for $\lambda_1 = -1$: $\mathbf{v}_1 = \begin{bmatrix} -r \\ 0 \\ r \end{bmatrix}, r \neq 0,$

for $\lambda_2 = 1$: $\mathbf{v}_2 = \begin{bmatrix} -s \\ 0 \\ 3s \end{bmatrix}, s \neq 0,$

for $\lambda_3 = 2$: $\mathbf{v}_3 = \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix}, t \neq 0$

$$C = \begin{bmatrix} -1 & -1 & 0 \\ 0 & 0 & 1 \\ 1 & 3 & 0 \end{bmatrix}, D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

9. Yes, the matrix is symmetric.

11. Yes, the eigenvalues are distinct.

13. F T T F T F F T F T

15. Assume that A is a square matrix such that $D = C^{-1}AC$ is a diagonal matrix for some invertible matrix C. Because $CC^{-1} = I$, we have $(CC^{-1})^T = (C^{-1})^TC^T = I^T = I$, so $(C^T)^{-1} = (C^{-1})^T$. Then $D^T = (C^{-1}AC)^T = C^TA^T(C^T)^{-1}$, so A^T is similar to the diagonal matrix D^T.

19. b. $A = O$ **c.** $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

21. $T([x, y]) = \dfrac{1}{1+m^2}[(1 - m^2)x + 2my, 2mx + (m^2 - 1)y]$

23. If A and B are similar square matrices, then $B = C^{-1}AC$ for some invertible matrix C. Let λ be an eigenvalue of A, and let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ be a basis for E_λ. Then Exercise 22 shows that $C^{-1}\mathbf{v}_1, C^{-1}\mathbf{v}_2, \ldots, C^{-1}\mathbf{v}_k$ are eigenvectors of B corresponding to λ, and Exercise 37 in Section 2.1 shows that these vectors are independent. Thus the eigenspace of λ relative to B has dimension at least as great as the eigenspace of λ relative to A. By the symmetry of the similarity relation (see Exercise 16), the eigenspace of λ relative to A has dimension at least as great as the eigenspace of λ relative to B. Thus the dimensions of these eigenspaces are equal—that is, the geometric multiplicity of λ is the same relative to A as relative to B.

31. Diagonalizable

33. Complex (but not real) diagonalizable

35. Complex (but not real) diagonalizable

37. Not diagonalizable

39. Real diagonalizable

41. Complex (but not real) diagonalizable

Section 5.3

1. a. $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 1 & 0 \end{bmatrix}$;

b. Neutrally stable;

c. $\begin{bmatrix} a_{k+1} \\ a_k \end{bmatrix} = A^k\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{2}{3}(1)^k\begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{3}\left(-\frac{1}{2}\right)^k \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. The sequence starts $0, 1, \frac{1}{2}, \frac{3}{4}, \frac{5}{8}$ and $A^3\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{2}{3}\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{24}\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{5}{8} \\ \frac{3}{4} \end{bmatrix}$, which checks.

d. For large k, we have $\begin{bmatrix} a_{k-1} \\ a_k \end{bmatrix} \approx \frac{2}{3}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \end{bmatrix}$, so $a_k \approx \frac{2}{3}$.

3. a. $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 1 & 0 \end{bmatrix}$; b. Neutrally stable;

c. $\begin{bmatrix} a_{k+1} \\ a_k \end{bmatrix} = A^k \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{3}(1)^k \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{3}\left(-\frac{1}{2}\right)^k \cdot$
$\begin{bmatrix} -1 \\ 2 \end{bmatrix}$. The sequence starts $1, 0, \frac{1}{2}, \frac{1}{4}, \frac{3}{8}$

and $A^3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{24}\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{3}{8} \\ \frac{1}{4} \end{bmatrix}$,

which checks.

d. For large k, we have $\begin{bmatrix} a_{k+1} \\ a_k \end{bmatrix} \approx \frac{1}{3}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$,

so $a_k \approx \frac{1}{3}$.

5. a. $A = \begin{bmatrix} 1 & \frac{3}{4} \\ 1 & 0 \end{bmatrix}$; b. Unstable;

c. $\begin{bmatrix} a_{k+1} \\ a_k \end{bmatrix} = A^k \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{4}\left(\frac{3}{2}\right)^k \begin{bmatrix} 3 \\ 2 \end{bmatrix} - \frac{1}{4}\left(-\frac{1}{2}\right)^k \cdot$
$\begin{bmatrix} -1 \\ 2 \end{bmatrix}$. The sequence starts $0, 1, 1, \frac{7}{4}, \frac{5}{2}$

and $A^3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{27}{32}\begin{bmatrix} 3 \\ 2 \end{bmatrix} + \frac{1}{32}\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{80}{32} \\ \frac{56}{32} \end{bmatrix}$

$= \begin{bmatrix} \frac{5}{2} \\ \frac{7}{4} \end{bmatrix}$, which checks.

d. For large k, we have $\begin{bmatrix} a_{k+1} \\ a_k \end{bmatrix} \approx \frac{1}{4}\left(\frac{3}{2}\right)^k \begin{bmatrix} 3 \\ 2 \end{bmatrix}$,

so $a_k \approx \frac{3^k}{2^{k+1}}$; and a_k approaches ∞ as k approaches ∞.

7. $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -k_1 e^{-3t} + 4k_2 e^{4t} \\ k_1 e^{-3t} + 3k_2 e^{4t} \end{bmatrix}$

9. $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2k_1 e^{t} + k_2 e^{4t} \\ k_1 e^{t} + k_2 e^{4t} \end{bmatrix}$

11. $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} k_1 e^{-3t} + k_2 e^{t} + k_3 e^{7t} \\ k_2 e^{t} + k_3 e^{7t} \\ k_2 e^{t} \end{bmatrix}$

13. $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -k_1 e^{-t} - k_2 e^{t} \\ k_3 e^{2t} \\ k_1 e^{-t} + 3k_2 e^{t} \end{bmatrix}$

CHAPTER 6

Section 6.1

1. $\frac{2}{5}[3, 4]$

3. $\mathbf{p}_1 = [1, 0, 0]$, $\mathbf{p}_2 = [0, 2, 0]$, $\mathbf{p}_3 = [0, 0, 1$

5. $\mathbf{p} = -\frac{1}{3}[2, -3, 1, 2]$

7. sp([1, 0, 1], [−2, 1, 0])

9. sp([−12, 4, 5])

11. sp([2, −7, 1, 0], [−1, −2, 0, 1])

13. a. $-5\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ b. $-5\mathbf{i} + 3\mathbf{j} + \mathbf{k}$

15. $\frac{1}{3}[5, 4, 1]$ **17.** $\frac{1}{7}[5, 3, 1]$

19. $\frac{1}{6}[2, -1, 5]$ **21.** $\frac{1}{3}[3, -2, -1, 1]$

23. F T T T F T T F F T **29.** $\frac{4}{5}$

31. $\sqrt{\frac{14}{5}}$ **33.** $\frac{\sqrt{161}}{3\sqrt{3}}$ **35.** $\sqrt{10}$ **37.**

Section 6.2

1. $[2, 3, 1] \cdot [-1, 1, -1] = -2 + 3 - 1 = ($
so the generating set is orthogonal.
$\mathbf{b}_W = \frac{1}{42}[136, 29, 103]$.

3. $[1, -1, -1, 1] \cdot [1, 1, 1, 1] =$
$1 - 1 - 1 + 1 = 0$,
$[1, -1, -1, 1] \cdot [-1, 0, 0, 1] =$
$-1 + 0 + 0 + 1 = 0$, and
$[1, 1, 1, 1] \cdot [-1, 0, 0, 1] =$
$-1 + 0 + 0 + 1 = 0$,
so the generating set is orthogonal; $\mathbf{b}_W =$
$[2, 2, 2, 1]$.

5. $\left\{\frac{1}{\sqrt{5}}[1, 0, -2], \frac{1}{\sqrt{70}}[6, -5, 3]\right\}$

7. $\left\{[0, 1, 0], \frac{1}{\sqrt{2}}[1, 0, 1]\right\}$

9. $\left\{\frac{1}{\sqrt{2}}[1, 0, 1], \frac{1}{\sqrt{3}}[-1, 1, 1], \frac{1}{\sqrt{6}}[1, 2, -\right.$

11. $\left\{\frac{1}{\sqrt{2}}[1, 0, 1, 0], [0, 1, 0, 0], \frac{1}{\sqrt{6}}[1, 0, -1\right.$

13. $\left[\frac{9}{2}, -3, \frac{9}{2}\right]$

15. $\left[\frac{4}{3}, 0, -\frac{1}{3}, \frac{5}{3}\right]$

17. $\left\{\frac{1}{\sqrt{2}}[1, 0, 1, 0], \frac{1}{\sqrt{6}}[-1, 2, 1, 0], \frac{1}{\sqrt{3}}[1, 1, -1, 0], [0, 0, 0, 1]\right\}$

19. $\{[3, -2, 0, 1], [-9, -8, 14, 11]\}$

21. $\left\{\frac{1}{\sqrt{6}}[2, 1, 1], \frac{1}{\sqrt{2}}[0, -1, 1]\right\}$

23. $\{[2, 1, -1, 1], [1, 1, 3, 0], [-24, 9, 5, 44]\}$

25. F T T F F T F T T T

27. $Q = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & 2/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{3} & -1/\sqrt{6} \end{bmatrix}$,

$R = \begin{bmatrix} \sqrt{2} & \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3} & -1/\sqrt{3} \\ 0 & 0 & 4/\sqrt{6} \end{bmatrix}$

33. $\left\{\sqrt{\frac{2}{\pi}} \sin x, \sqrt{\frac{2}{\pi}} \cos x\right\}$

35. $\left\{1, \sqrt{\frac{2}{4e - e^2 - 3}}\,(e^x - e + 1)\right\}$

Section 6.3

1. Let A be the given matrix. Then

$$A^TA = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

so A is orthogonal and $A^{-1} = A^T = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$.

3. Let A be the given matrix. Then

$$A^TA = \frac{1}{7}\begin{bmatrix} 2 & 3 & -6 \\ -3 & 6 & 2 \\ 6 & 2 & 3 \end{bmatrix} \frac{1}{7}\begin{bmatrix} 2 & -3 & 6 \\ 3 & 6 & 2 \\ -6 & 2 & 3 \end{bmatrix} = \frac{1}{49}\begin{bmatrix} 49 & 0 & 0 \\ 0 & 49 & 0 \\ 0 & 0 & 49 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

so A is orthogonal and $A^{-1} = A^T = \frac{1}{7}\begin{bmatrix} 2 & 3 & -6 \\ -3 & 6 & 2 \\ 6 & 2 & 3 \end{bmatrix}$.

5. $\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{6} & \frac{1}{6} \end{bmatrix}$

7. $\frac{1}{49}\begin{bmatrix} 1 & \frac{3}{2} & -3 \\ -3 & 6 & 2 \\ 6 & 2 & 3 \end{bmatrix}$

9. $\pm\frac{1}{\sqrt{6}}\begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$

11. $\pm\frac{\sqrt{23}}{6}$

13. $\frac{1}{\sqrt{2}}\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$

15. $\frac{1}{2}\begin{bmatrix} 1 & -2 & 1 \\ -\sqrt{2} & 0 & \sqrt{2} \\ 1 & 2 & 1 \end{bmatrix}$

17. $\begin{bmatrix} 0 & \frac{1}{2} & -1\sqrt{2} & \frac{1}{2} \\ -1/\sqrt{2} & -\frac{1}{2} & 0 & \frac{1}{2} \\ 1/\sqrt{2} & -\frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 1/\sqrt{2} & \frac{1}{2} \end{bmatrix}$

19. F T T T T T T T F T

23. $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

27. An orthogonal matrix A gives rise to an orthogonal linear transformation $T(\mathbf{x}) = A\mathbf{x}$ that preserves the magnitude of vectors. Thus, if $A\mathbf{v} = \lambda\mathbf{v}$, so that $T(\mathbf{v}) = \lambda\mathbf{v}$, we must have $\|\mathbf{v}\| = \|\lambda\mathbf{v}\| = |\lambda|\|\mathbf{v}\|$. If $\mathbf{v}$ is an eigenvector, so that $\mathbf{v} \neq \mathbf{0}$, it follows that $|\lambda| = 1$; so $\lambda = \pm 1$.

33. No

35. Yes

37. Yes

Section 6.4

1. $P = \frac{1}{6}\begin{bmatrix} 4 & 2 & -2 \\ 2 & 1 & -1 \\ -2 & -1 & 1 \end{bmatrix}$, projection $= \frac{1}{2}\begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$

3. $P = \frac{1}{35}\begin{bmatrix} 34 & -3 & 5 \\ -3 & 26 & 15 \\ 5 & 15 & 10 \end{bmatrix}$, projection $= \frac{1}{35}\begin{bmatrix} 86 \\ 13 \\ 25 \end{bmatrix}$

5. $P = \frac{1}{6}\begin{bmatrix} 5 & -1 & 2 \\ -1 & 5 & 2 \\ 2 & 2 & 2 \end{bmatrix}$, projection $= \frac{1}{3}\begin{bmatrix} 2 \\ 8 \\ 5 \end{bmatrix}$

7. $P = \frac{1}{21}\begin{bmatrix} 10 & -1 & 3 & 10 \\ -1 & 19 & 6 & -1 \\ 3 & 6 & 3 & 3 \\ 10 & -1 & 3 & 10 \end{bmatrix}$, projection $= \frac{1}{21}\begin{bmatrix} 41 \\ 40 \\ 27 \\ 41 \end{bmatrix}$

9. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ **11.** $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

13. If P is the projection matrix for a subspace W of $\mathbb{R}^n$ and if $\mathbf{b} \in \mathbb{R}^n$ is a column vector, then the projection of $\mathbf{b}$ on W is $P\mathbf{b}$. Because $P\mathbf{b}$ is in W, geometry indicates that the projection of $P\mathbf{b}$ on W is again $P\mathbf{b}$. Thus, $P(P\mathbf{b}) = P\mathbf{b}$, so $P^2\mathbf{b} = P\mathbf{b}$ and $(P^2 - P)\mathbf{b} = \mathbf{0}$ for all $\mathbf{b} \in \mathbb{R}^n$. It follows from Exercise 41 in Section 1.3 that $P^2 - P = O$, so $P^2 = P$.

15. F T T F F T F F T T **17.** I

19. a. 0, 1

b. 0 has geometric and algebraic multiplicity $n - k$,

1 has geometric and algebraic multiplicity k.

c. Because the algebraic and geometric multiplicities of each eigenvalue are equal, P is a diagonalizable matrix.

21. The $n \times n$ identity matrix I for each positive integer n.

23. $\begin{bmatrix} \frac{9}{25} & \frac{12}{25} & 0 \\ \frac{12}{25} & \frac{16}{25} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

25. $\dfrac{1}{49}\begin{bmatrix} 13 & -18 & 0 & -12 \\ -18 & 36 & 12 & 0 \\ 0 & 12 & 13 & -18 \\ -12 & 0 & -18 & 36 \end{bmatrix}$

27. $\dfrac{1}{3}\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & -1 & 1 \\ 0 & -1 & 2 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix}$ **29.** $\begin{bmatrix} 10 \\ -4 \\ -2 \end{bmatrix}$ **31.** $\begin{bmatrix} 14 \\ 0 \\ -7 \\ 14 \end{bmatrix}$

33. Referring to Figure 6.11, we see that, for $\mathbf{p} = P\mathbf{b}$, the vector from the tip of $\mathbf{b}$ to the tip of $\mathbf{p}$ is $\mathbf{p} - \mathbf{b}$, which is also the vector from the tip of $\mathbf{p}$ to the tip of $\mathbf{b}_r$. Thus, the vector $\mathbf{b}_r = \mathbf{b} + 2(\mathbf{p} - \mathbf{b}) = 2\mathbf{p} - \mathbf{b} = 2(P\mathbf{b}) - \mathbf{b} = (2P - I)\mathbf{b}$.

35. The projections are approximately $\begin{bmatrix} 1.151261 \\ -1.184874 \\ 3.89916 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 3 \\ -1 \end{bmatrix}$, and $\begin{bmatrix} 1.932773 \\ -.806723 \\ 4.378151 \end{bmatrix}$, respectively.

37. The projections are approximately $\begin{bmatrix} 1.864516 \\ 1.496774 \\ -.135484 \\ 2.819355 \end{bmatrix}$, $\begin{bmatrix} 1.058064 \\ .787097 \\ -.941936 \\ 2.077419 \end{bmatrix}$, and $\begin{bmatrix} 4.116129 \\ 2.574194 \\ 1.116129 \\ 3.154839 \end{bmatrix}$ respectively.

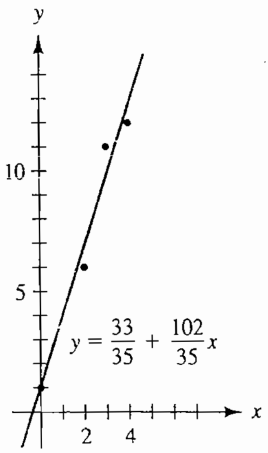
Section 6.5

1. a. $y = \dfrac{116.4}{59} + \dfrac{60.4}{59}x$

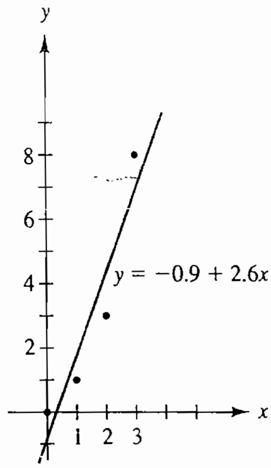
b. $\approx$ 7.092 inches

3. a. $y = .528e^{.274x}$ **b.** $\approx$ \$27,

5. $y = \dfrac{33}{35} + \dfrac{102}{35}x$



7. $y = -0.9 + 2.6x$



9. $y = 0.1 - 0.4x + x^2$

11. $y = 1.6 + 2x$

13. 4.5 min

15. Let $t = x - c$, where $c = (\Sigma_{i=1}^{m} a_i)/m$. The data points $(a_1 - c, b_1), (a_2 - c, b_2), \ldots, (a_m - c, b_m)$ have the property that $\Sigma_{i=1}^{m} (a_i - c) = 0$. Exercise 14 then shows that these data points have least-squares linear fit given by $y = r_0 + r_1 t$, where r_0 and r_1 have the values given in Exercise 14. Making the substitution $t = x - c$, we see that the data points $(a_1, b_1), (a_2, b_2), \ldots, (a_m, b_m)$ have the least-squares linear fit given by $y = r_0 + r_1(x - c)$.

17. $\bar{x} = \begin{bmatrix} -\frac{1}{5} \\ \frac{3}{5} \end{bmatrix}$

19. $\bar{x} = \begin{bmatrix} 0 \\ 2 \\ -\frac{1}{4} \end{bmatrix}$

21. F F T T F F T T F F

23. See answer to Exercise 17.

25. See answer to Exercise 19.

27. The computer gave the fit $y = 0.7587548 + 1.311284x$ with a least-squares sum of 0.03891051.

29. We achieved a least-squares sum of 5.838961 with the expoential fit $y = 0.8e^{0.2x}$. The computer achieved a least-squares sum of 6.34004 with the exponential fit $y = 0.8874836e^{0.1960377x}$. The fit using logarithms tries to fit the smaller y-value data accurately at the expense of the larger y-value data, so that the *percent* accuracy of fit to the y-coordinates is as good as possible.

31. The computer gave the fit $y = 12.03846 - 1.526374x$ with a least-squares sum of 0.204176.

33. $y \approx 5.476 - 0.75x + 0.2738x^2$

35. $y \approx 5.632 - 1.139x + 0.1288x^2 + 0.05556x^3 + 0.01512x^4$

37. $y = -5 - 8x + 9x^2 - x^3$

HAPTER 7

ction 7.1

1. $[-1, 1]$

3. $[-4, -2, 1, 5]$

5. $[3, 5, 1, 1]$

7. $2x^2 + 6x + 2$

9. $\begin{bmatrix} -1 \\ -4 \\ -2 \end{bmatrix}$

11. a. $C_{B,B'} = \begin{bmatrix} 2 & 1 & 2 \\ 3 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$;

b. $C_{B',B} = \begin{bmatrix} -6 & 3 & 4 \\ 9 & -4 & -6 \\ 2 & -1 & -1 \end{bmatrix}$

13. a. $C_{B,B'} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$;

b. $C_{B',B} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & -1 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix}$

15. $C_{B',B} = \begin{bmatrix} 1 & 2 & 1 \\ -1 & -2 & -2 \\ 0 & 1 & 0 \end{bmatrix}$

17. $C_{B',B} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$

19. $C_{B,B'} = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 1 & -1 & 0 & 1 \\ 0 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

21. $C_{B,B'} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$

23. T F T F T F T F T T

25. $C_{B,B''} = C_{B',B} C_{B,B'}$

Section 7.2

1. $R_B = \begin{bmatrix} 6 & 7 \\ -3 & -3 \end{bmatrix}$, $R_{B'} = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}$, $C = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$

3. $R_B = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $R_{B'} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 0 \end{bmatrix}$

5. $R_B = \begin{bmatrix} \frac{13}{5} & \frac{4}{5} & 2 \\ -\frac{11}{5} & -\frac{3}{5} & -2 \\ -\frac{9}{5} & -\frac{2}{5} & -2 \end{bmatrix}$, $R_{B'} = \begin{bmatrix} -\frac{4}{3} & -\frac{1}{6} & -\frac{10}{3} \\ -\frac{4}{3} & -\frac{5}{3} & -\frac{16}{3} \\ 1 & \frac{1}{2} & 3 \end{bmatrix}$,

$C = \begin{bmatrix} 0 & -\frac{9}{5} & -\frac{8}{5} \\ 1 & \frac{13}{5} & \frac{21}{5} \\ 0 & \frac{12}{5} & \frac{14}{5} \end{bmatrix}$

7. $R_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$, $R_{B'} = \frac{1}{3}\begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{bmatrix}$,

$C = \frac{1}{3}\begin{bmatrix} 1 & 1 & -2 \\ 1 & -2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

9. $R_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $R_{B'} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$,

$C = \frac{1}{2}\begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 2 & 0 \end{bmatrix}$

11. $R_B = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 1 & 1 & 2 \end{bmatrix}$, $R_{B'} = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{bmatrix}$,

$C = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

13. $R_B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, $R_{B'} = \begin{bmatrix} 0 & 1 & -2 & -3 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$,

$C = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

15. $\begin{bmatrix} 2 & 1 & -3 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

17. $\lambda_1 = -1$, $\lambda_2 = 5$; $E_{-1} = \mathrm{sp}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$,

$E_5 = \mathrm{sp}\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right)$; diagonalizable

19. $\lambda_1 = 0$, $\lambda_2 = 1$, $\lambda_3 = 2$; $E_0 = \mathrm{sp}\left(\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}\right)$,

$E_1 = \mathrm{sp}\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right)$, $E_2 = \mathrm{sp}\left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right)$; diagonalizable

21. $\lambda_1 = -2$, $\lambda_2 = \lambda_3 = 5$; $E_{-2} = \mathrm{sp}\left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right)$,

$E_5 = \mathrm{sp}\left(\begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix}\right)$; not diagonalizable

23. F T T F T T T F F T

CHAPTER 8

Section 8.1

1. $U = \begin{bmatrix} 3 & -6 \\ 0 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 3 & -3 \\ -3 & 1 \end{bmatrix}$

3. $U = \begin{bmatrix} 1 & -4 & 3 \\ 0 & -1 & -8 \\ 0 & 0 & 0 \end{bmatrix}$, $A = \begin{bmatrix} 1 & -2 & \frac{3}{2} \\ -2 & -1 & -4 \\ \frac{3}{2} & -4 & 0 \end{bmatrix}$

5. $U = \begin{bmatrix} -2 & 8 \\ 0 & 3 \end{bmatrix}$, $A = \begin{bmatrix} -2 & 4 \\ 4 & 3 \end{bmatrix}$

7. $U = \begin{bmatrix} 8 & 5 & -4 \\ 0 & 1 & -2 \\ 0 & 0 & 10 \end{bmatrix}$, $A = \begin{bmatrix} 8 & \frac{5}{2} & -2 \\ \frac{5}{2} & 1 & -1 \\ -2 & -1 & 10 \end{bmatrix}$

9. $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}\begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$, $-t_1^2 + t_2^2$

11. $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3/\sqrt{10} & -1/\sqrt{10} \\ 1/\sqrt{10} & 3/\sqrt{10} \end{bmatrix}\begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$, $-t_1^2 +$

13. $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}\begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$, $t_1^2 + 5t_2^2$

15. $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \end{bmatrix}\begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix}$,

$-t_1^2 + 2t_2^2 + 2t_3^2$

17. $a + c = k$, $ac = b^2$

19. $-5.472136t_1^2 + 3.472136t_2^2$

21. $-4.021597t_1^2 + 1.323057t_2^2 + 4.69854$

23. $-4t_1^2 + \frac{1}{2}t_2^2 + 4t_3^2 + \frac{11}{2}t_4^2$